

- QQ group: student number + name
- Theory problems - Theory course TA (QQ group name before have remark) + teacher
- Experiment problems - Experiment course TA + teacher
- Classroom problem and international student– TA (along with the class)
- Hard to solve - Guo



《CAD与工程制图》

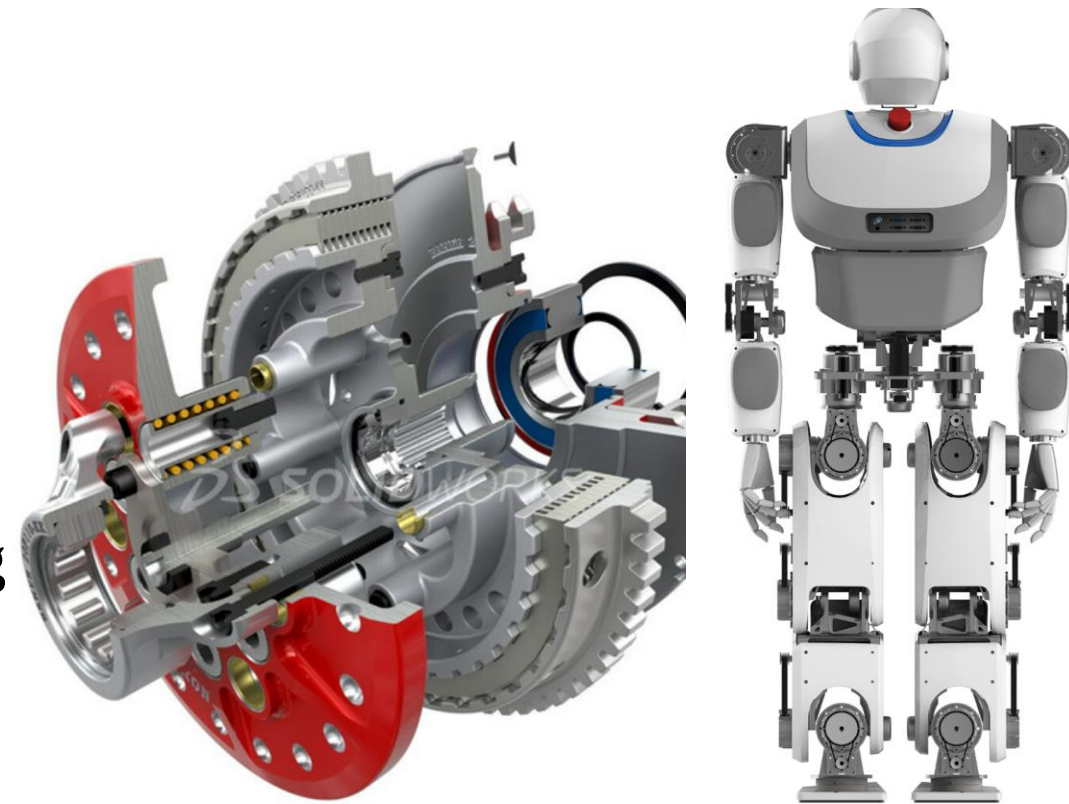
CAD (Computer Aided Design) and Engineering Drawing

2024年 秋季

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Southern University of Science and Technology



Chapter 3 Projections of Points, Lines, and Planes

- **3-1 Projection of Points**
- **3-2 Projection of Lines**
- **3-3 Projection of Planes**

Learning Objectives

- Develop a **sense of three-dimensional spatial perception** of objects
- Understand the basic principles of projecting **three-dimensional space** onto a **two-dimensional plane**

Chapter 3 Projections of Points, Lines, and Planes

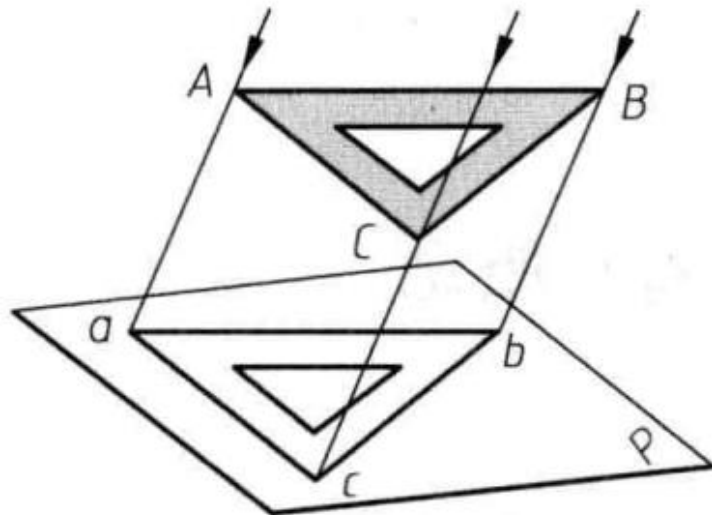
- **3-1 Projection of Points**
- **3-2 Projection of Lines**
- **3-3 Projection of Planes**

1、Classification of Projections

Parallel Projection Lines and Projection Plane

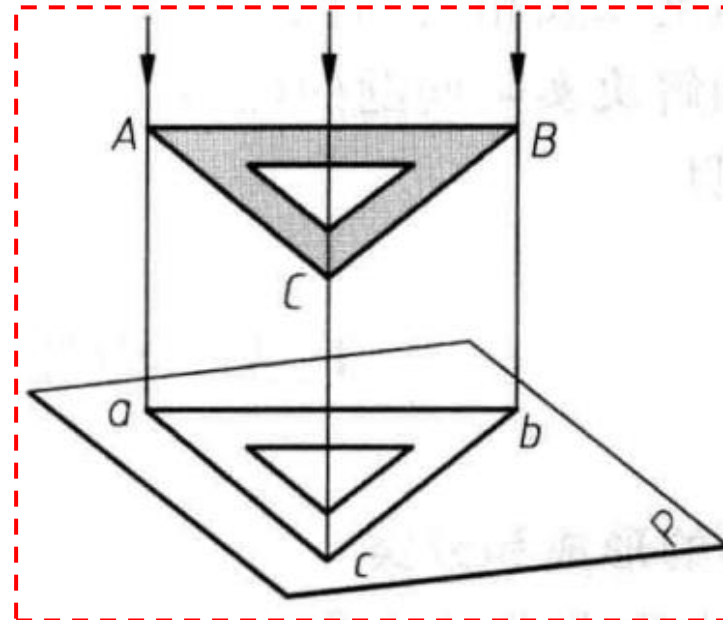
Non-perpendicular

Oblique Projection



Perpendicular

Orthographic Projection



Advantages: Accurate shape representation, easy drawing, good measurability. **Widely used in mechanical engineering** and key to this course.

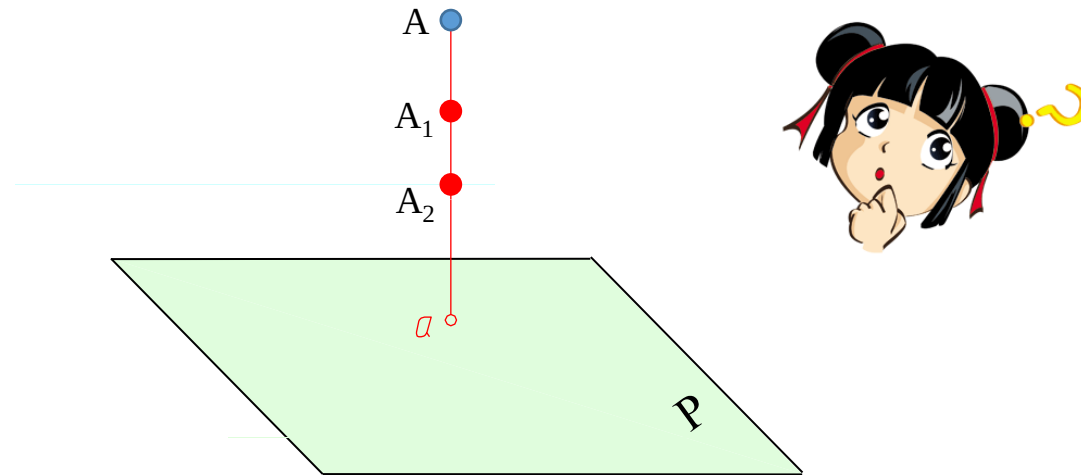
Disadvantages: Poor 3D effect. **Requires engineering drawing knowledge and spatial imagination.**

Can a projection in one direction express the shape and size of an object?

How many directions of projections are needed at least?



Projection of a Point in a Single Projection Plane System



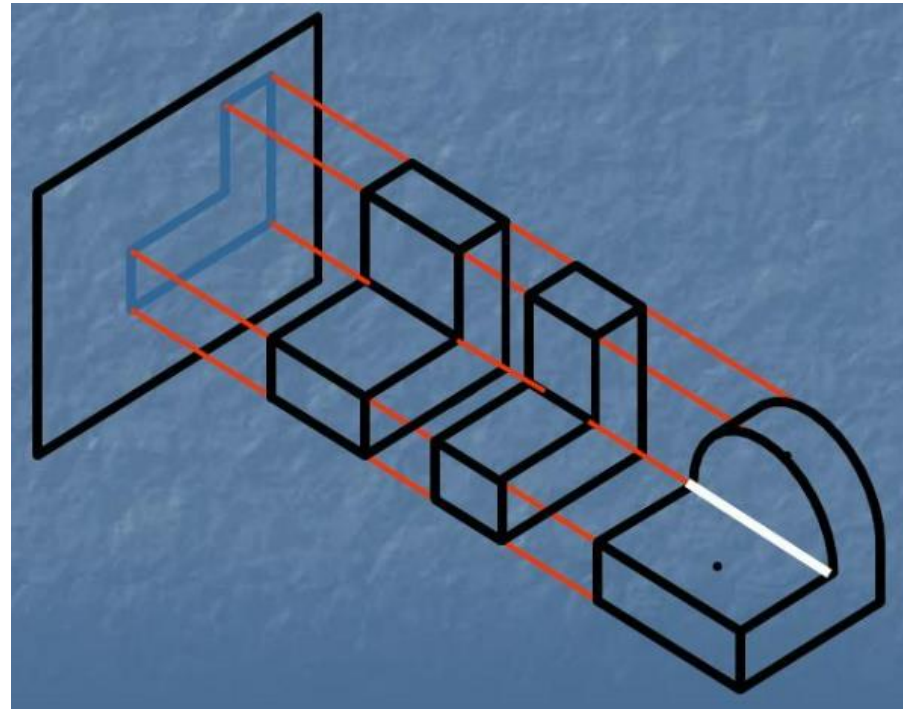
Can the projection of a point on one projection plane determine the spatial position of the point?

No

How to uniquely determine the spatial position of a point?

Multi-surface orthographic projection

A single orthographic projection cannot fully determine the object.



Two directions of projections can determine the size of the object, but **another direction of projection** is needed to determine the shape of the object.

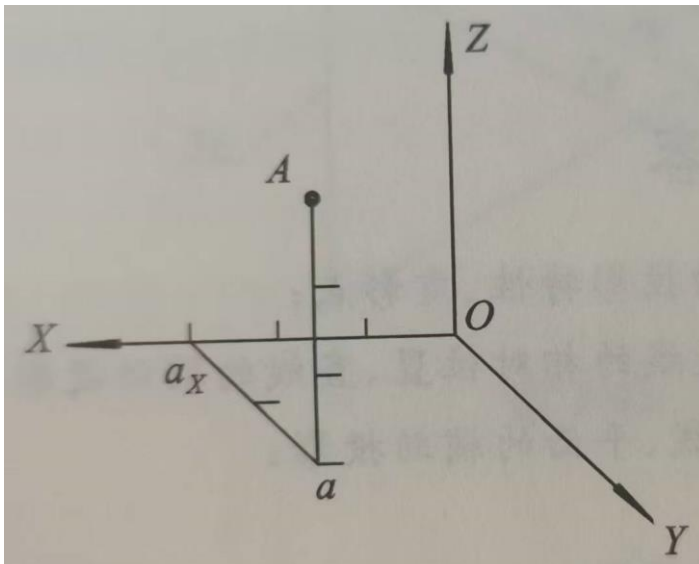
3-1 Projection of a Point



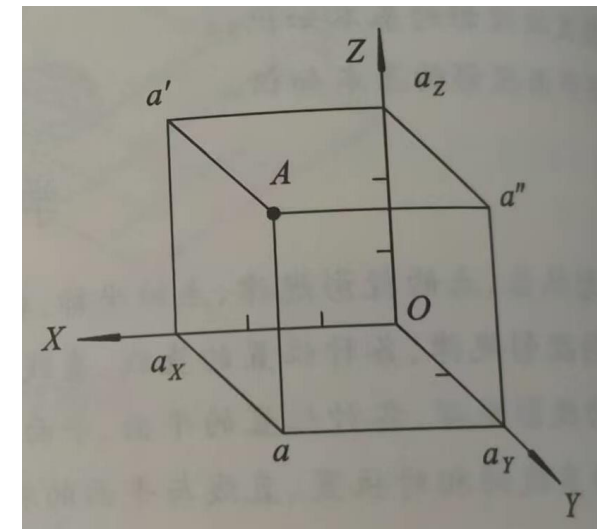
1. Point in a Rectangular Coordinate System

If the coordinates of point A are known to be (X_A, Y_A, Z_A) , there are two methods to determine its spatial position:

(1) Move successively along the X, Y, and Z axes by the coordinate values.



(2) Locate three points on the X, Y, and Z axes, draw parallel lines to form a hexahedron, and the vertex is point A.



3-1 Projection of a Point

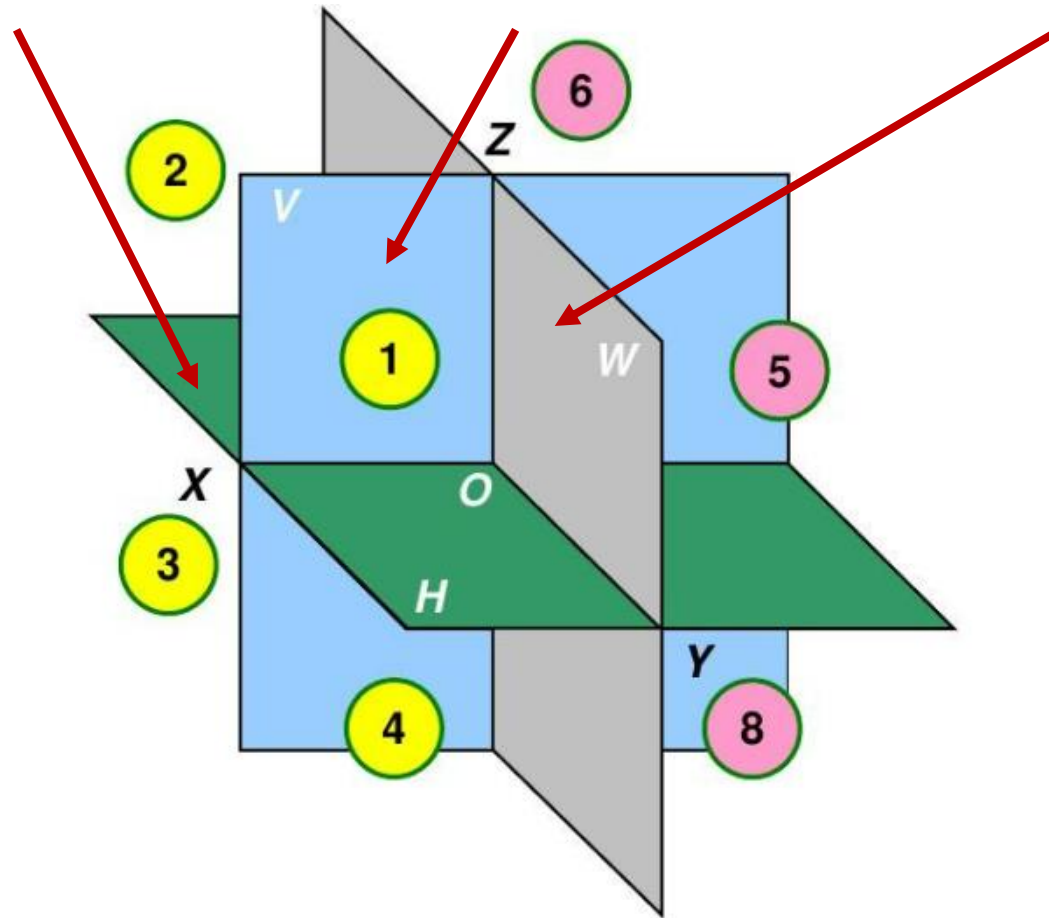


2. Three-Projection Plane System

H: Horizontal Projection
Plane

V: Frontal Plane

W: Profile Plane



Three Projection Planes
Eight Views

3-1 Projection of a Point

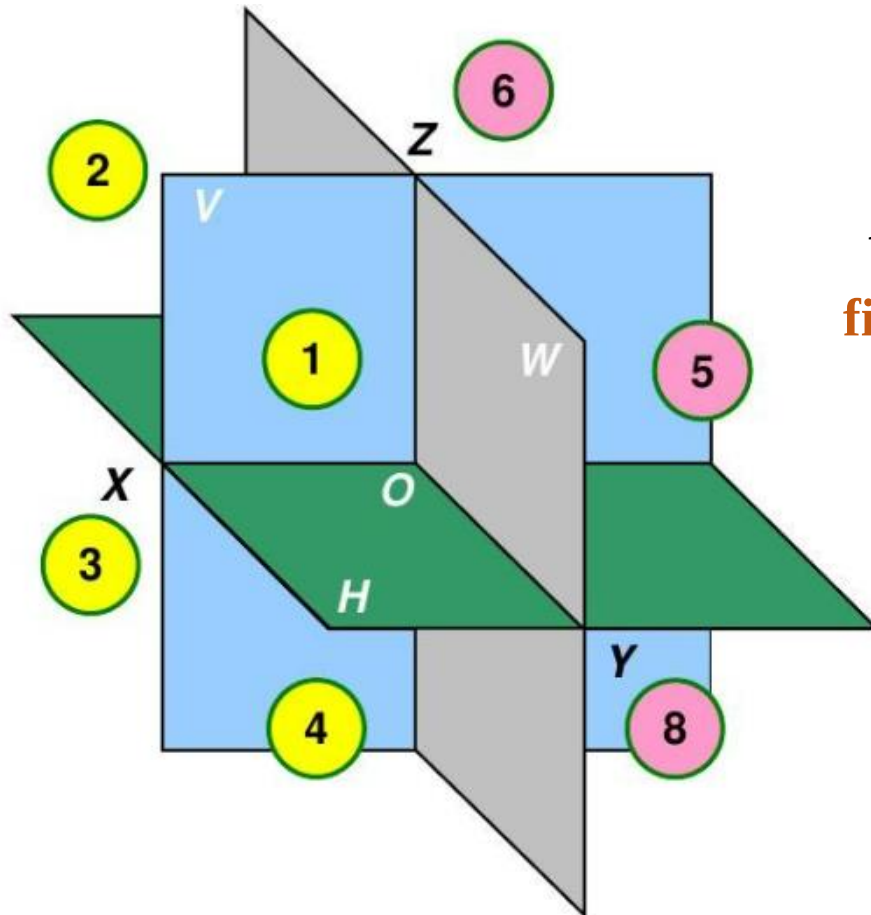


2. Three-Projection Plane System

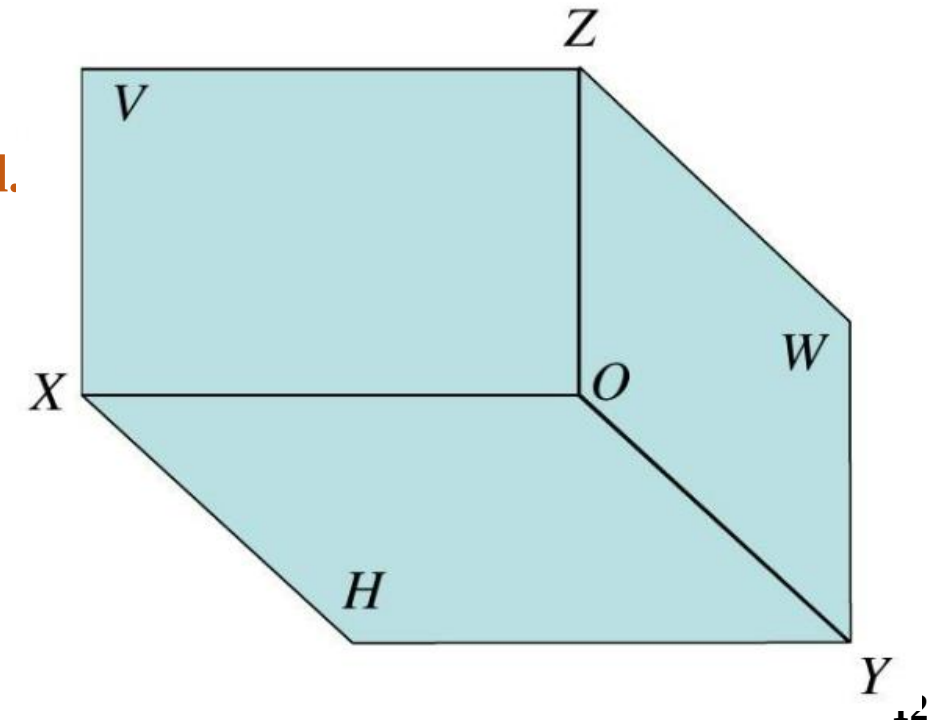
H: Horizontal Projection Plane

V: Frontal Plane

W: Profile Plane



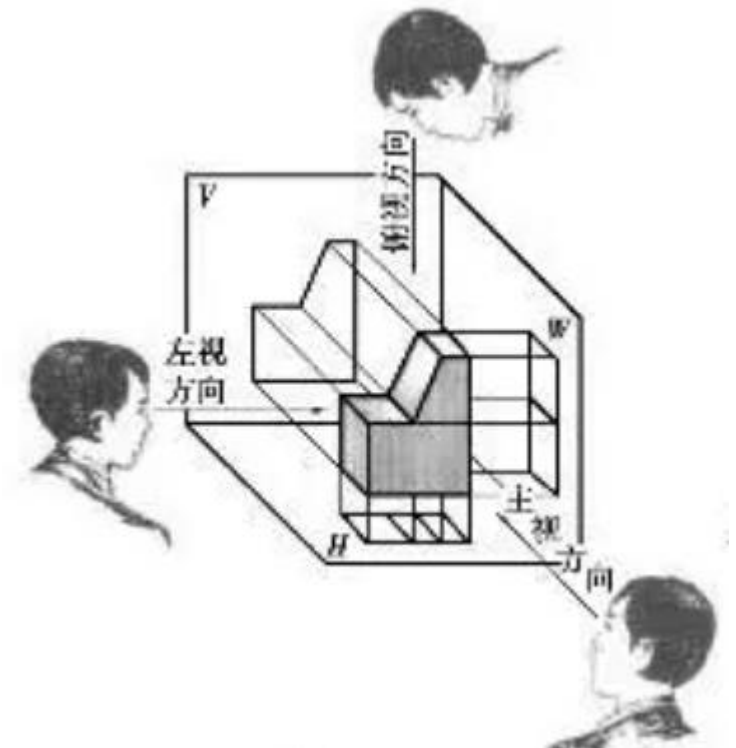
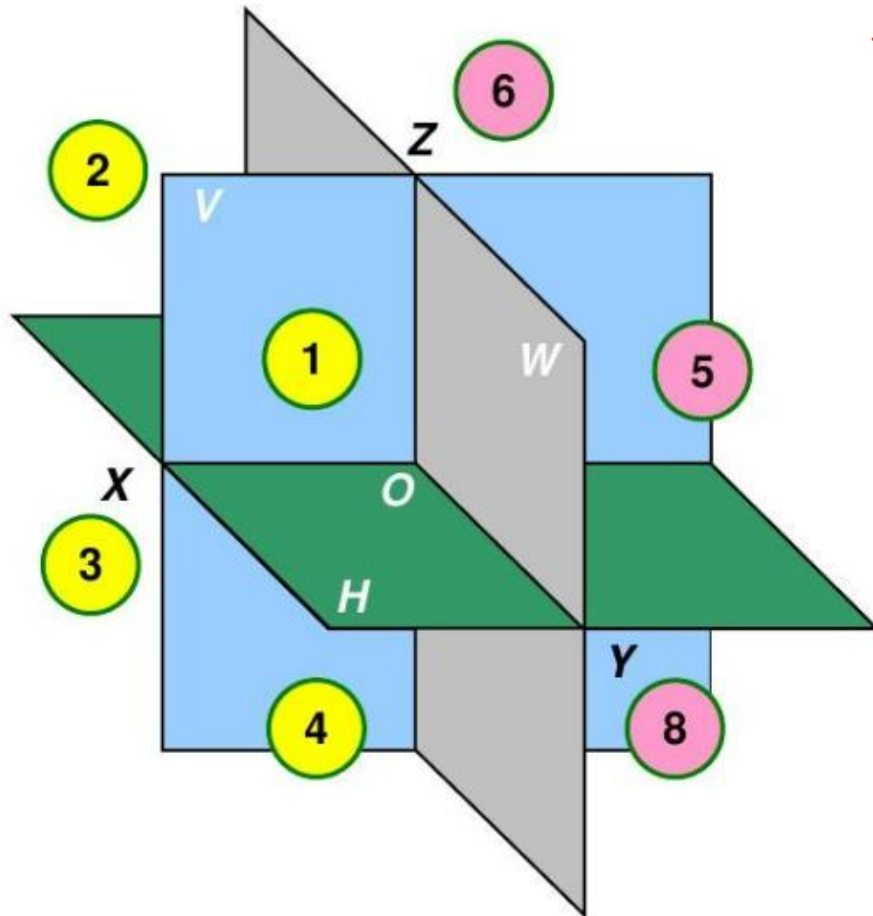
As per national standards, the first angle projection is adopted.



3-1 Projection of a Point



Why is the first angle
projection selected?



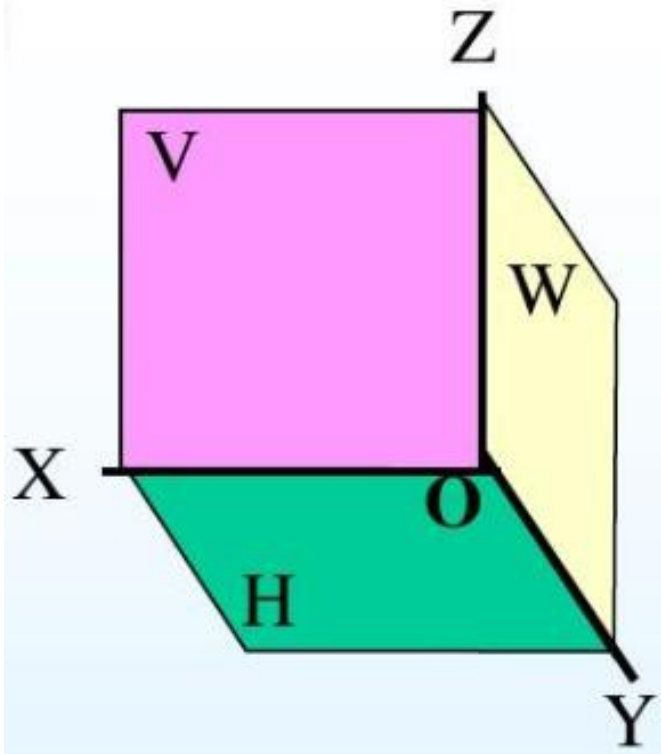
3-1 Projection of a Point



2. Three-Projection Plane System

H: Horizontal Projection Plane
V: Frontal Plane

W: Profile Plane

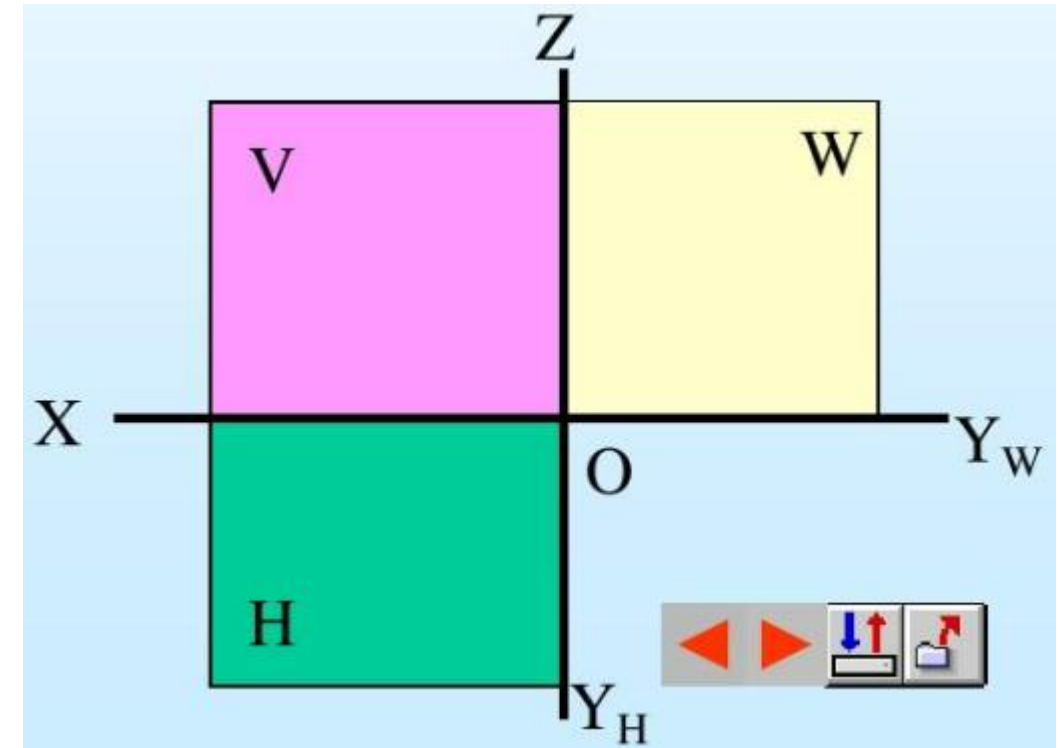


Specifications:

- The front plane V remains stationary.
- The horizontal plane H rotates 90° downward around the OX axis.
- The side plane W rotates 90° to the right around the OZ axis.



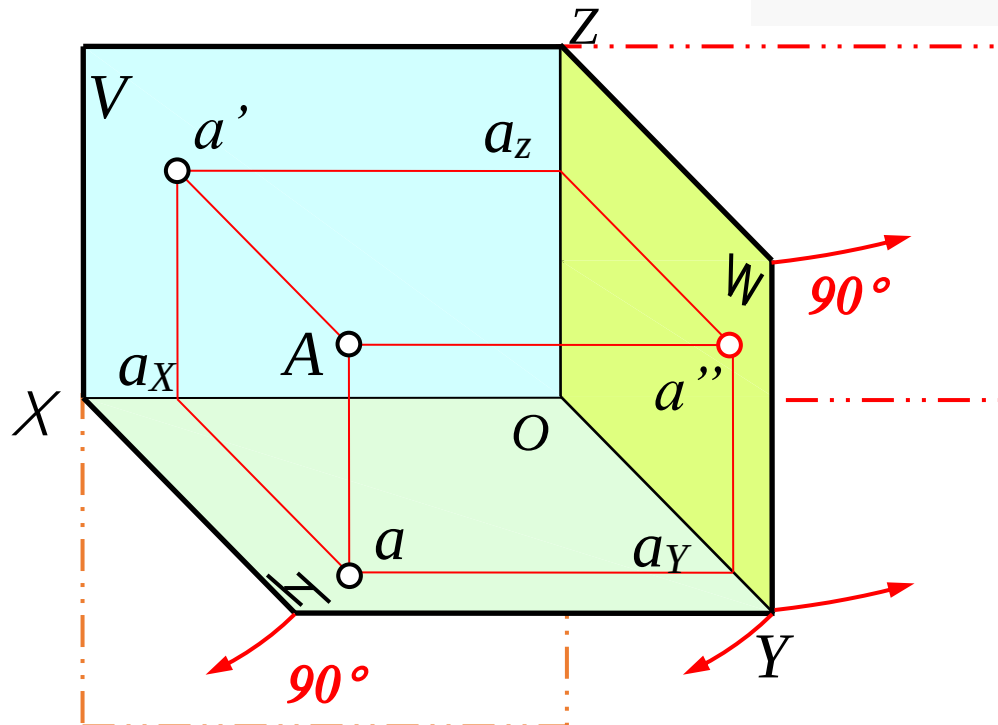
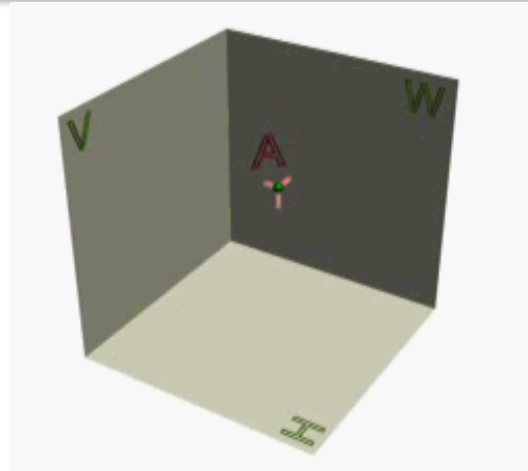
Expansion



3-1 Projection of a Point



3. Tri-Projection of a Point



Unfolding of
Projection
Planes



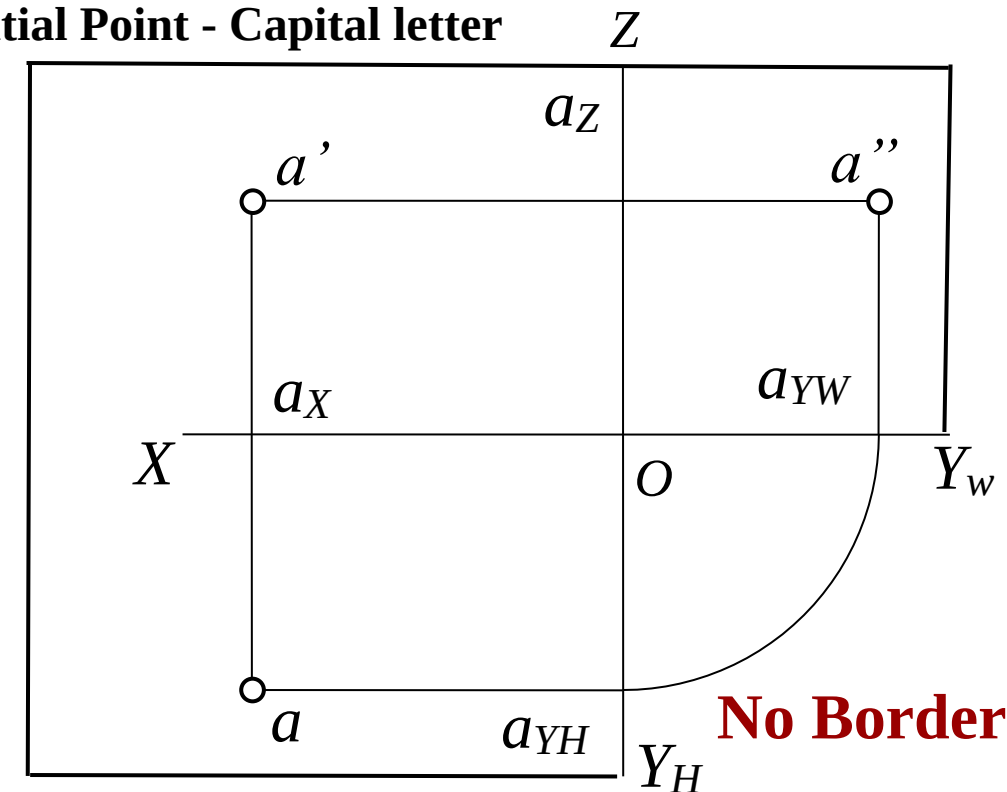
Symbol Conventions:

H Projection - Lowercase letter

V Projection - Lowercase letter with an
apostrophe

W Projection - Lowercase letter with two
apostrophes

Spatial Point - Capital letter

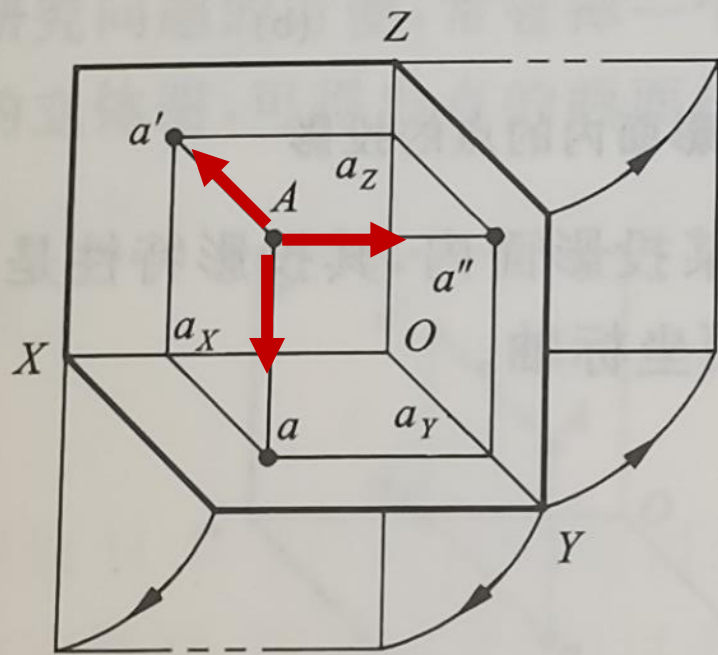


3-1 Projection of a Point

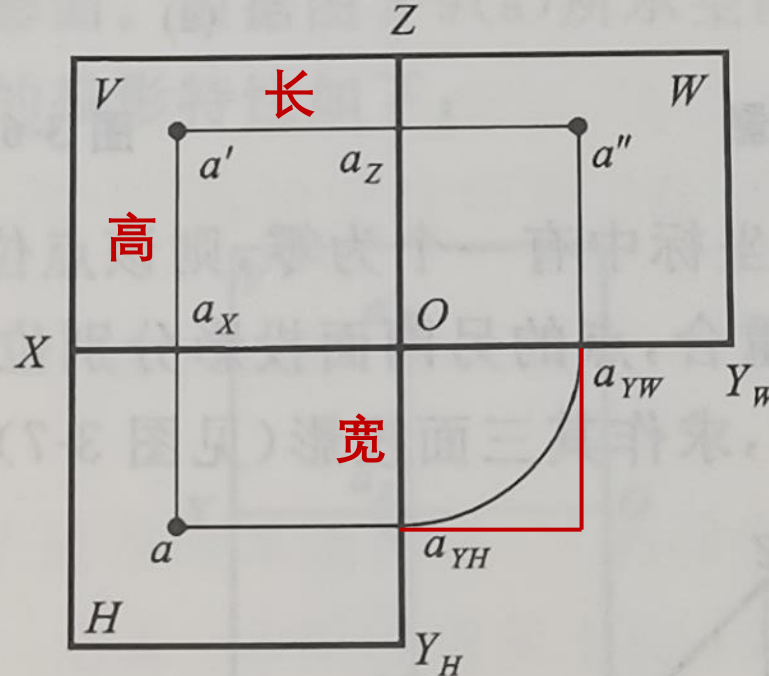


3. Tri-Projection of a Point

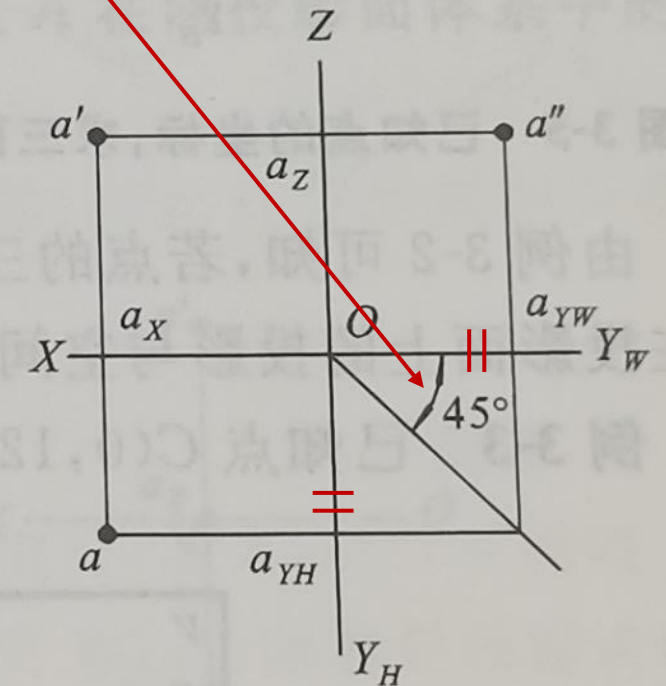
Why is this angle 45° ?



(a) Axonometric
Drawing



(b) Unfold
Projection Planes



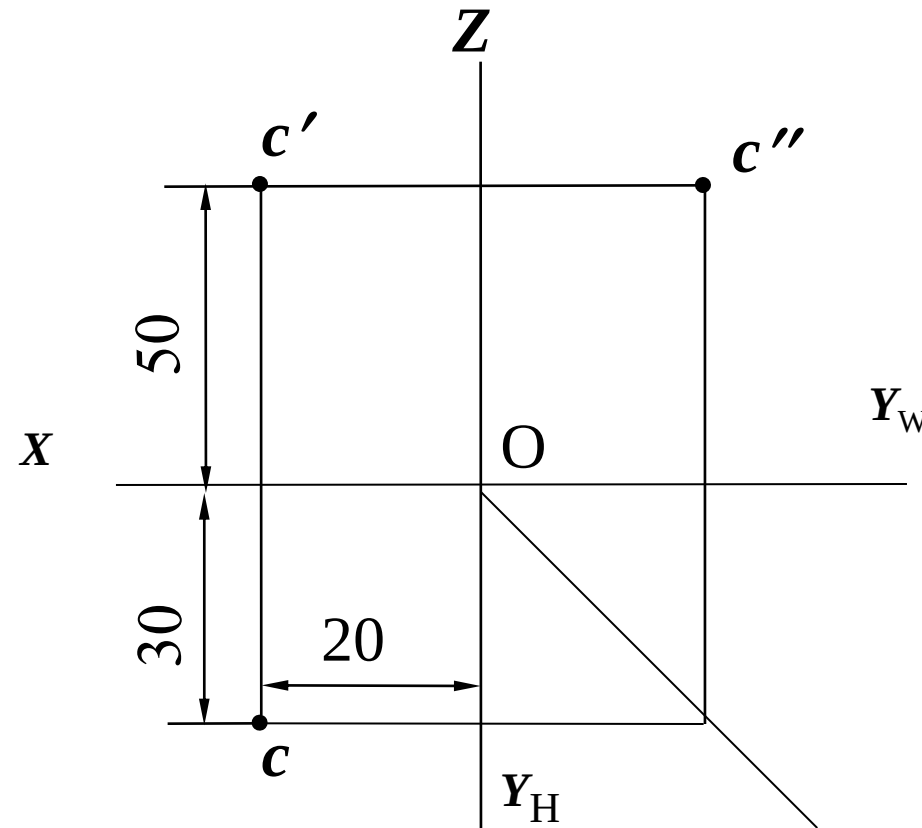
(c) Projection
Drawing

3-1 Projection of a Point



3. Tri-Projection of a Point

Example 1: Given the coordinates of point C (20, 30, 50), find the projections of point C. (Draw the diagram)

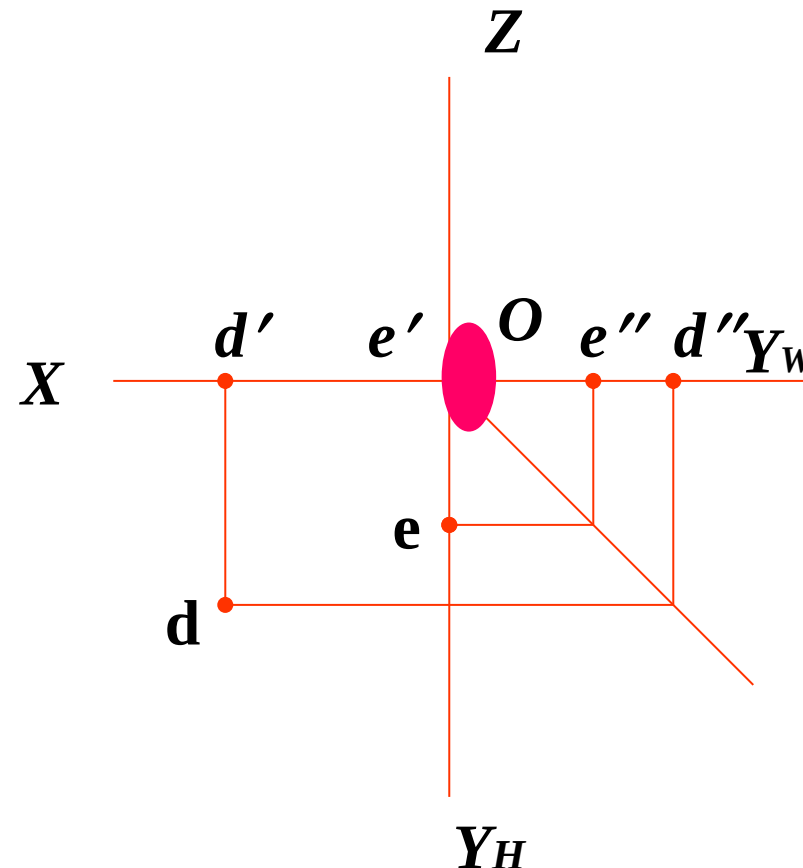


3-1 Projection of a Point



3. Tri-Projection of a Point

Example 2: Given the coordinates of point D (40, 40, 0) and point E (0, 20, 0), find the projections of points D and E. (Draw the diagram)

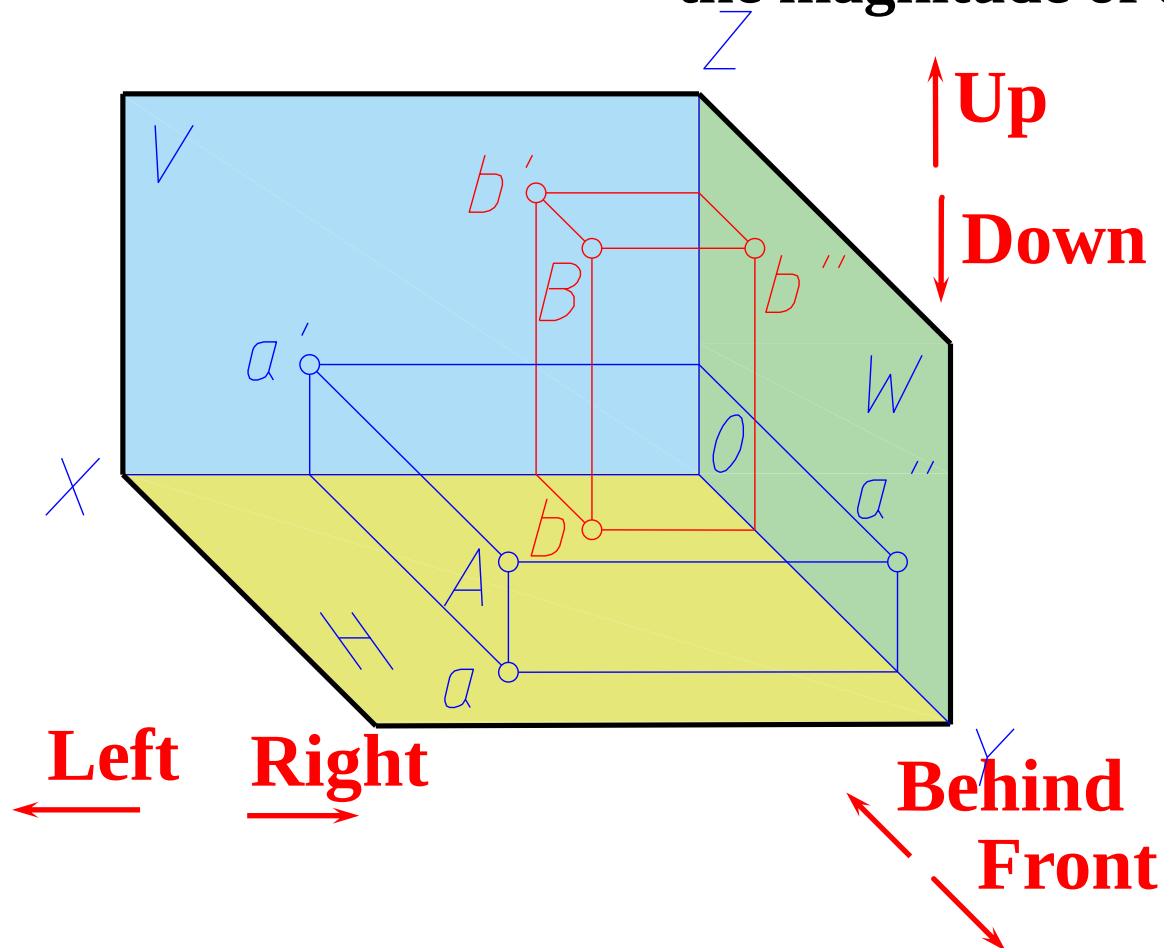


3-1 Projection of a Point



4. Relative Position of Points

Determine the relative up/down, front/back, left/right positions of two points based on the magnitude of their coordinates



Convention:

X-axis direction is referred to as left and right.

Y-axis direction is referred to as front and back.

Z-axis direction is referred to as up and down.

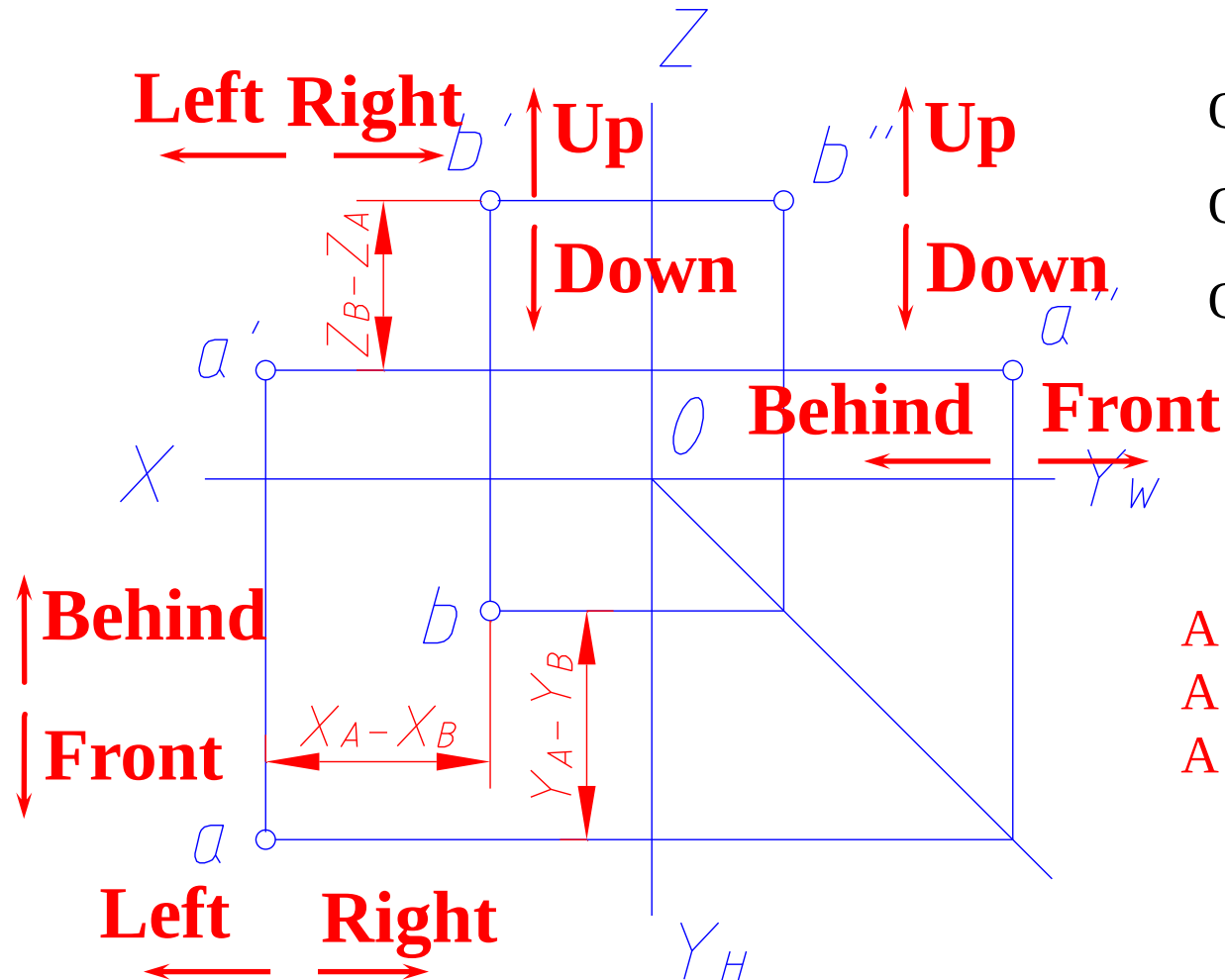
Point B is to the right, behind, and above point A.

3-1 Projection of a Point



Method for determining the relative position of two points

■ Determine the relative position relationship of two points on the projection drawing.



On the H plane: Left-right, Front-back

On the V plane: Left-right, Up-down

On the W plane: Up-down, Front-back

Point A is to the left, front, and below point B.

A point with a larger x-coordinate is on the left.

A point with a larger y-coordinate is in front.

A point with a larger z-coordinate is above.

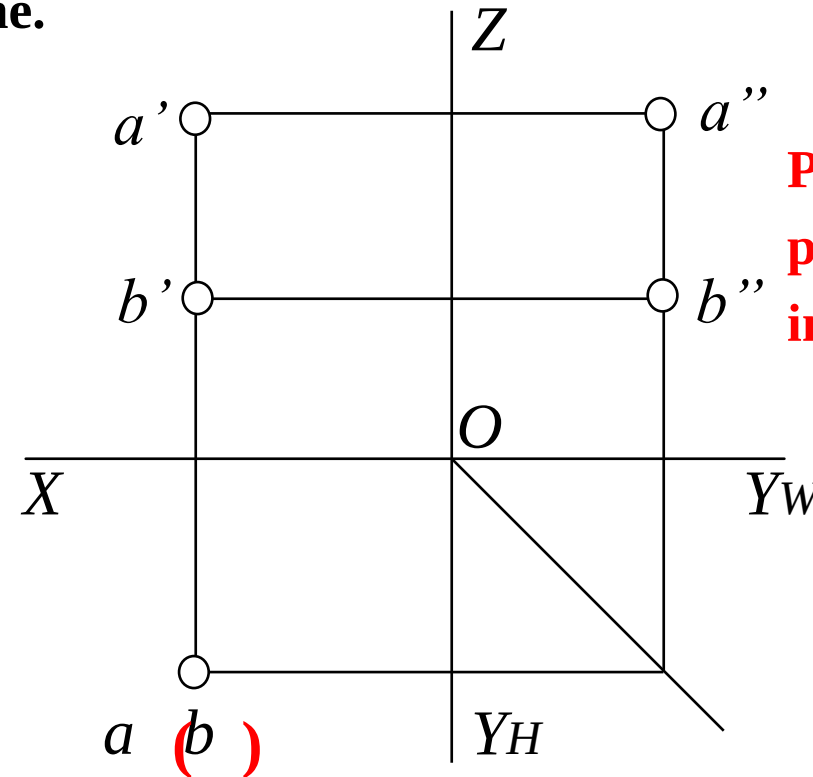
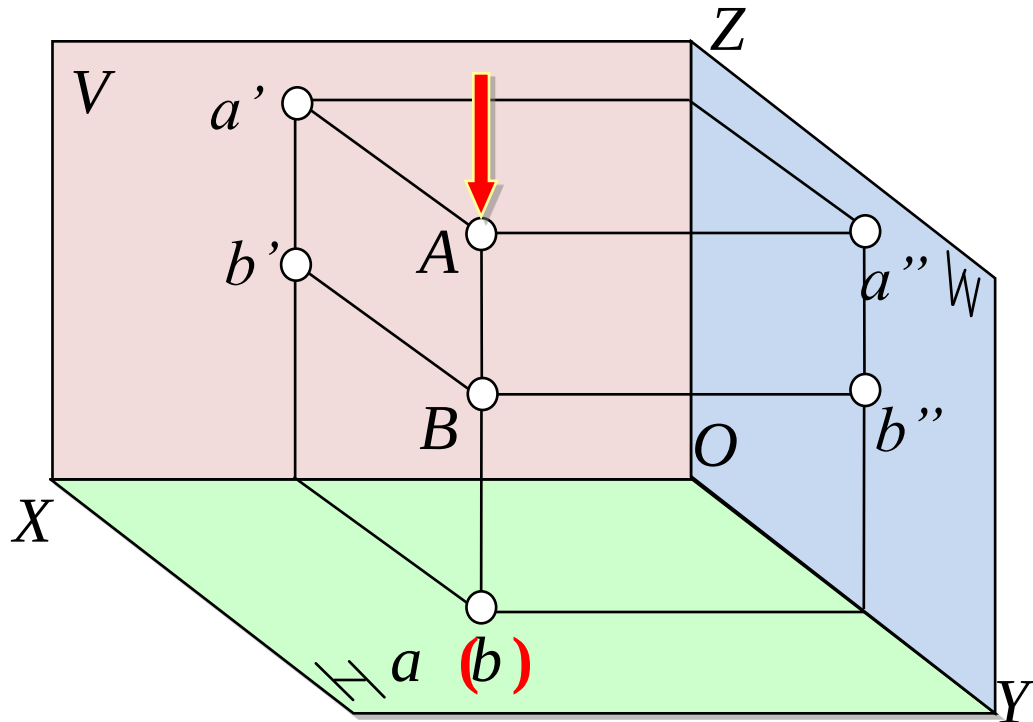
3-1 Projection of a Point



4. Relative Position of Points - Shadow Points

Two points are said to be **shadow points** with respect to a projection plane if their projections coincide at a single point on that plane.

Points A and B are shadow points on the H plane.



Points with invisible
projections are enclosed
in parentheses

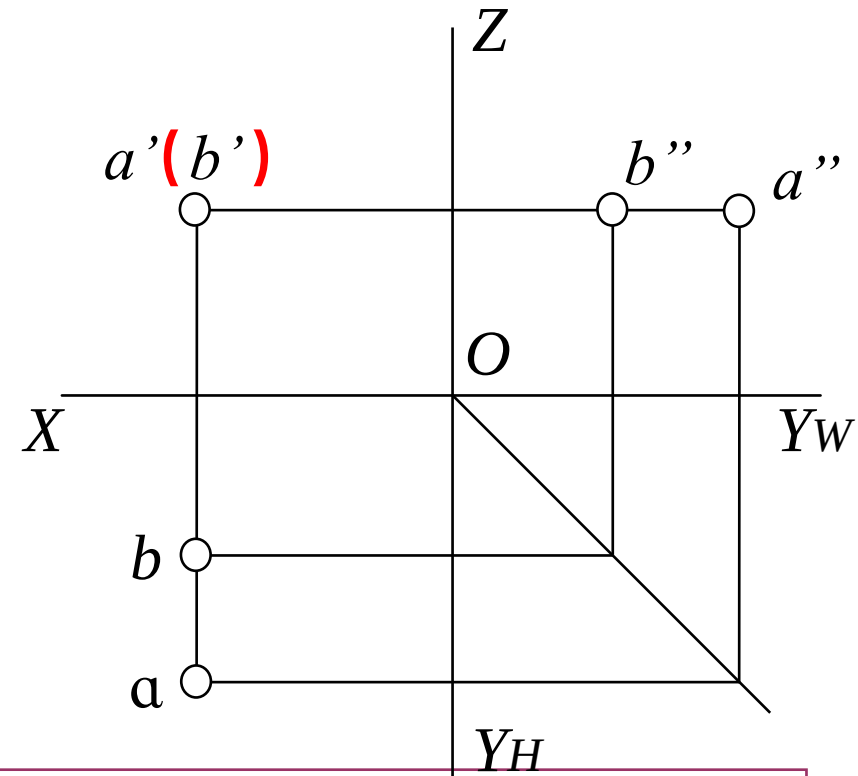
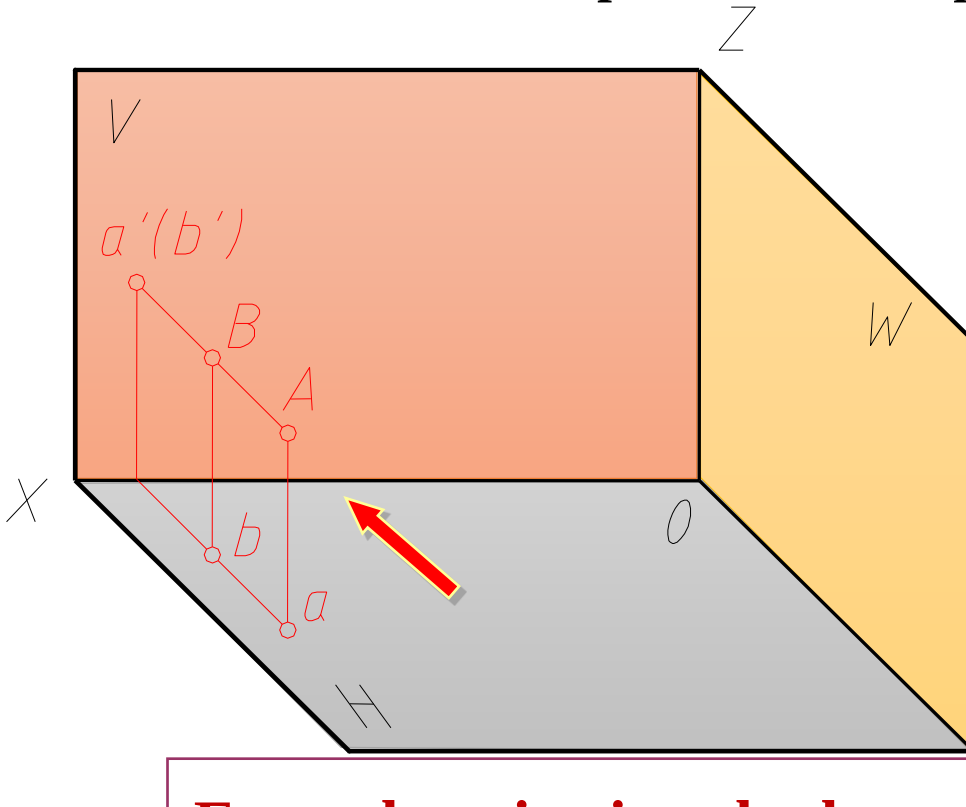
Horizontal projection shadow; determine the up-down relationship from the front (side) projection.

3-1 Projection of a Point



4. Relative Position of Points - Shadow Points

Points A and B are shadow points on the V plane.



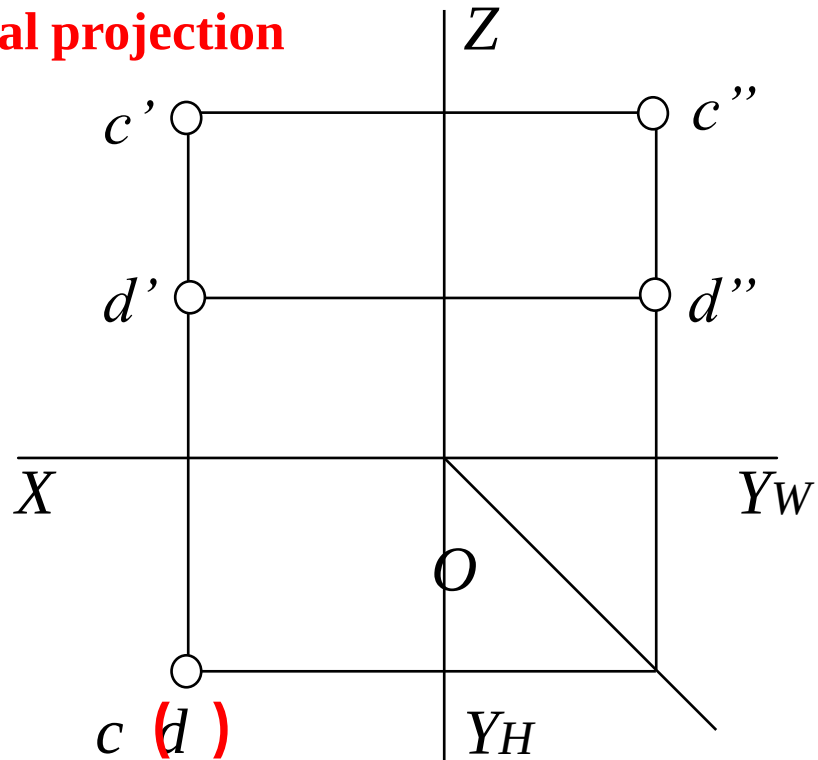
Frontal projection shadow; determine the front-back relationship from the horizontal (and side) projection

3-1 Projection of a Point



Summary: Visibility determination of shadow points.

Horizontal projection shadow

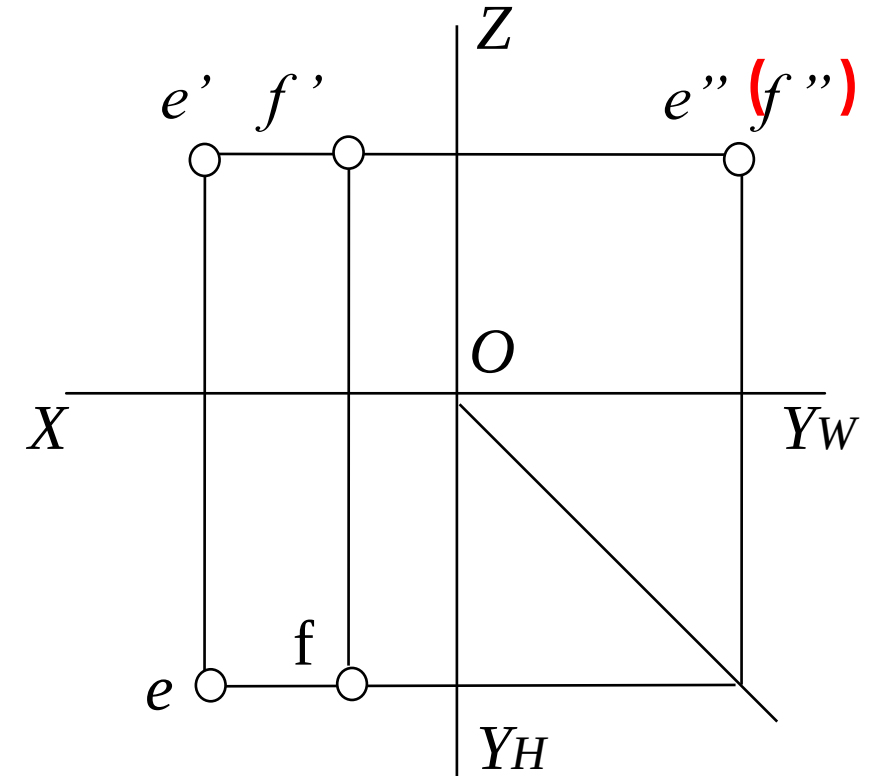


Determine up-down from front (and side) projections:

Higher Z value is above, visible;

Lower Z value is below, invisible.

Side projection shadow



Determine front-back from horizontal plane (and front) projections:

- Larger X value is in front, visible;

- Smaller X value is behind, invisible.

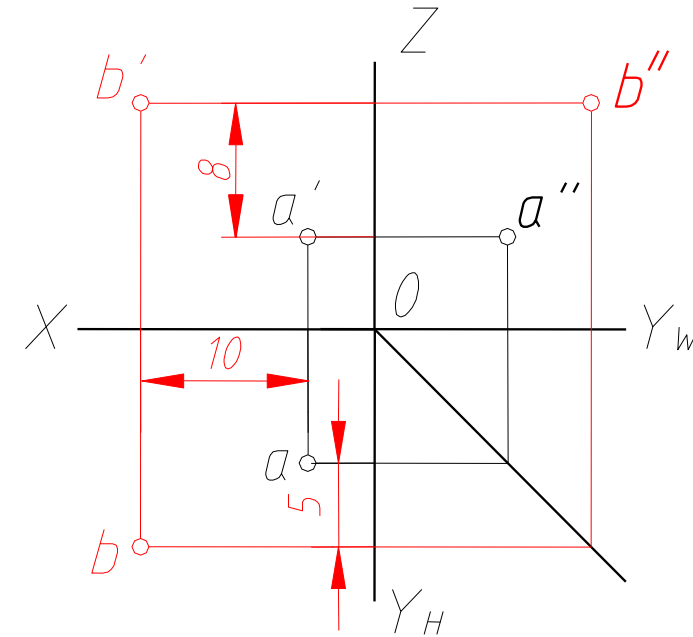
3-1 Projection of a Point



Example 2-3: Given the tri-projection of point A as $\alpha, \alpha', \alpha''$, and knowing that point B is 10mm to the left, 8mm above, and 5mm in front of point A, find the tri-projection of point B as b, b', b'' .

Analysis:

Based on the relationship between the relative positions of projections of two points in space and their coordinates, using point A as a reference, $x_B - x_A = 10$, $y_B - y_A = 5$, $z_B - z_A = 8$.



3-1 Projection of a Point



5. Bi-Projection of a Point

Two-plane projection system: Frontal projection and horizontal projection

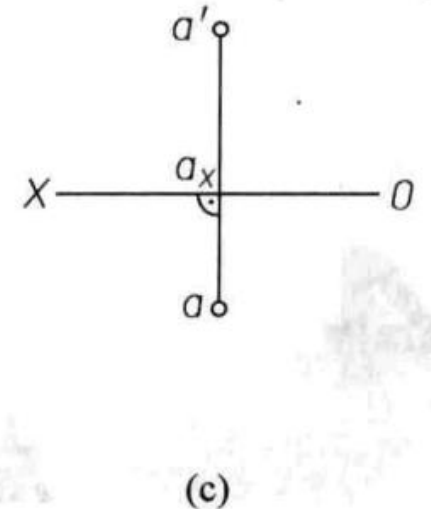
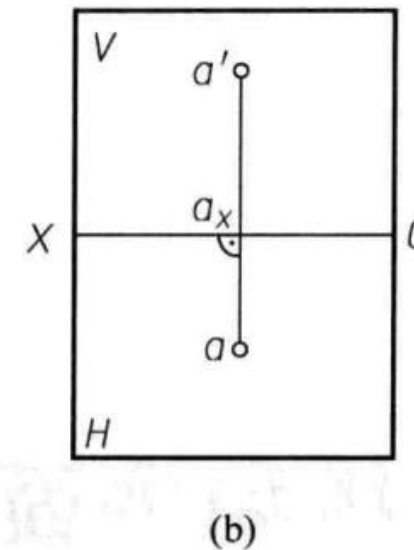
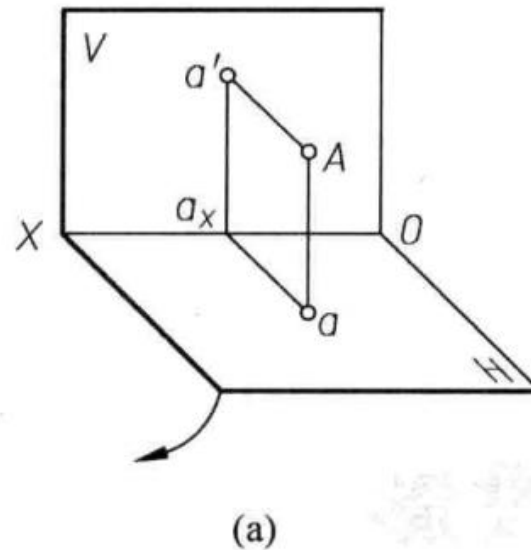
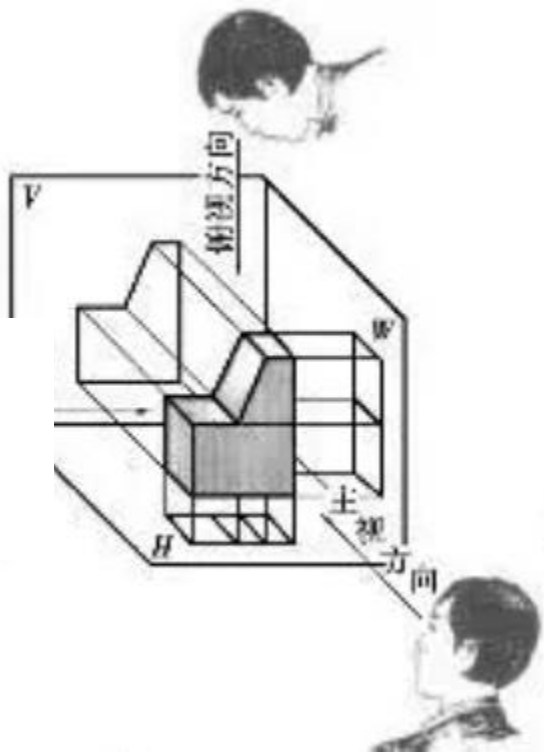


图 4-2 点在两投影面体系中的投影
Figure 4-2 Projection of a point in a two-plane system

Chapter 3 Projections of Points, Lines, and Planes

- **3-1 Projection of Points**
- **3-2 Projection of Lines**
- **3-3 Projection of Planes**

1. Projection of a Line

How to draw the projection of line segment AB?

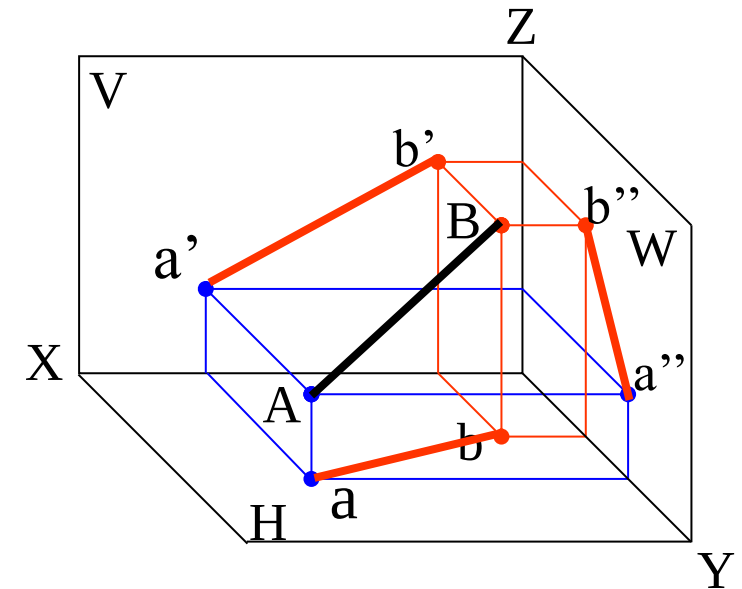
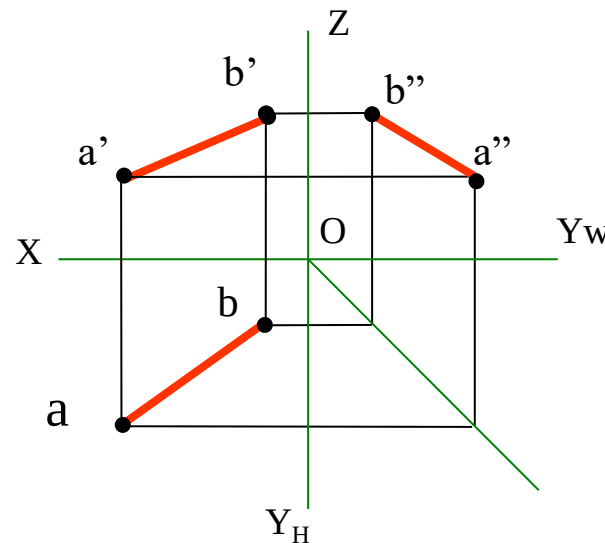
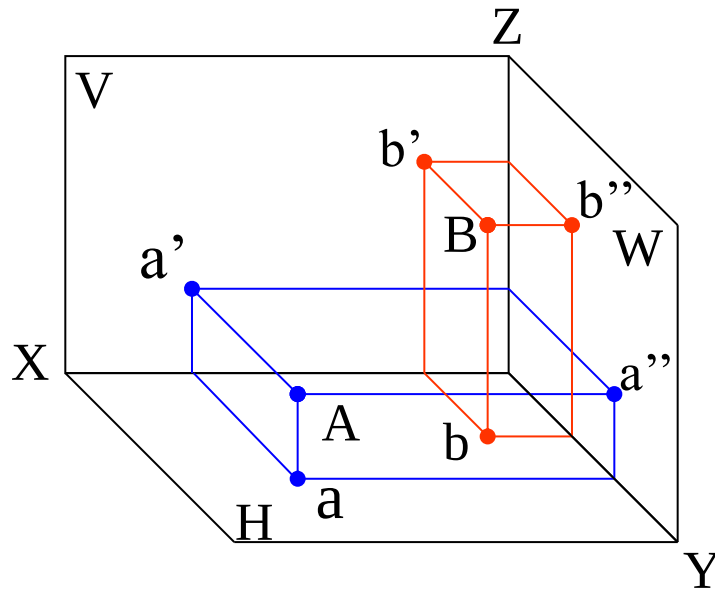


3-2 Projection of Lines



1. Projection of a Line

Two points determine a straight line, and connecting the projections of any two points on the line gives the projection of the line.

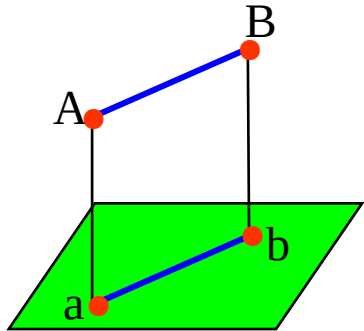


3-2 Projection of Lines



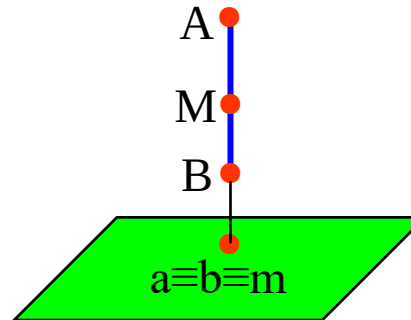
2. Characteristics of the Projection of a Line on a Projection Plane

There are three relative positions of a line to a projection plane:



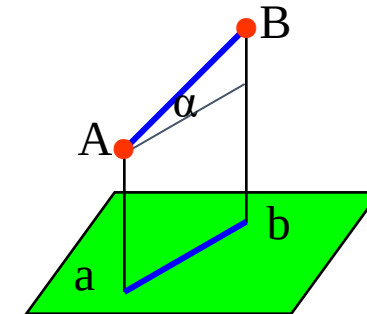
The line is parallel to the projection plane.
The projection reflects the true length of the line segment.
 $ab = AB$

True length



Line perpendicular to the projection plane.
Projection condenses to a single point.

Gathering



Line inclined to the projection plane.
Projection is shorter than the spatial line segment.
 $ab = AB \cos \alpha$

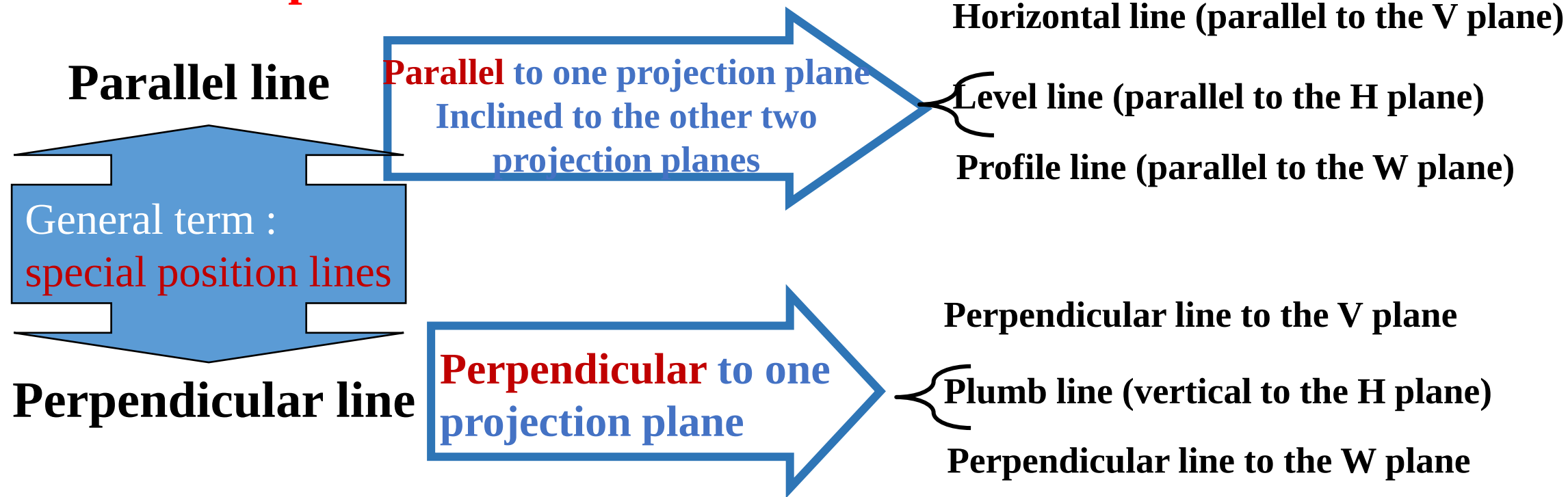
Contraction

3-2 Projection of Lines



3. Characteristics of Line Projections on Three Projection Planes

Projection characteristics of lines in various positions

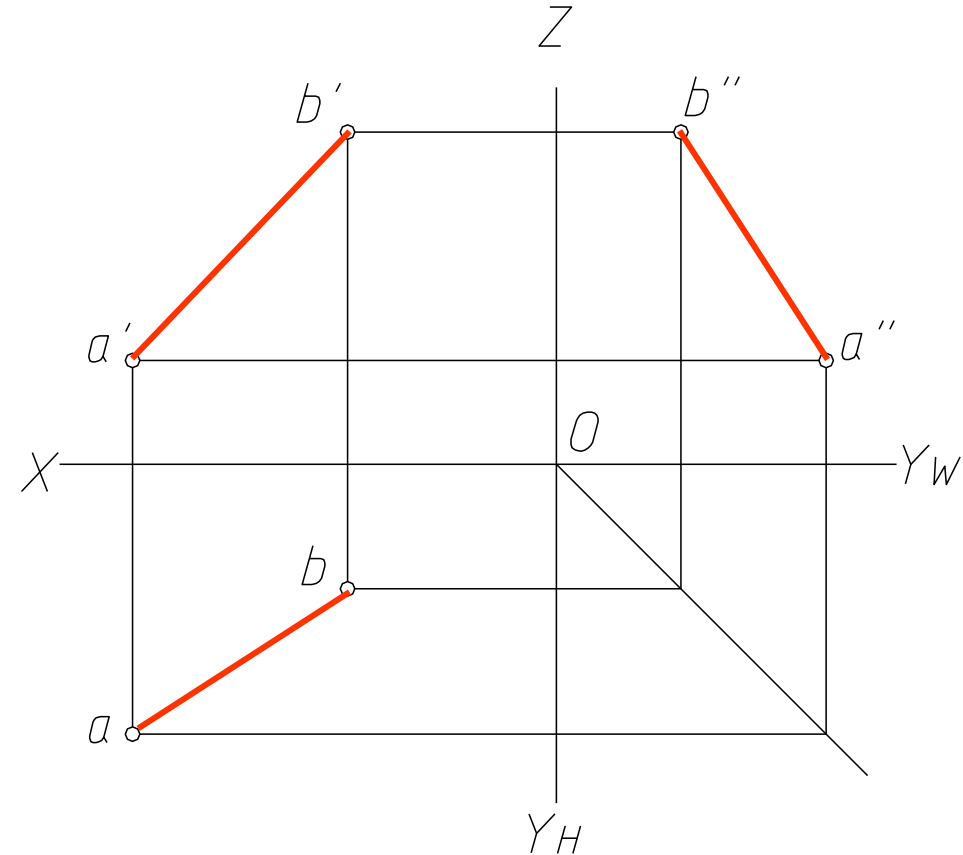
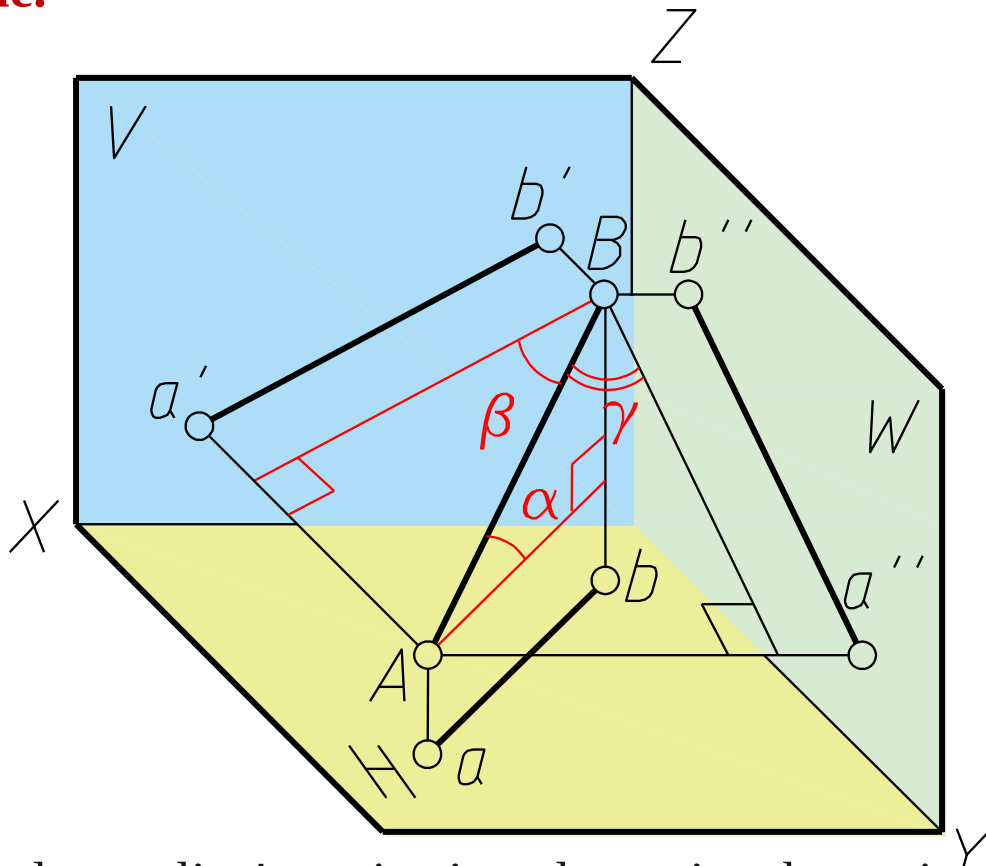


General position line A line inclined to all three projection planes

3-2 Projection of Lines



General position line: A line inclined to all three projection planes is called a **general position line**.



To draw a line's projection, determine the projections of its endpoints and connect these points in same plane.

Projection Characteristics: All three projections are inclined line segments, and each is shorter than the actual length of the line segment.

3-2 Projection of Lines



(I) Parallel to Projection Plane

Projection Characteristics:

- The projection of the line on the **parallel** projection plane reflects **the true length**, and the **angle** between the projection and the corresponding axis reflects the actual angle between the line and the other two projection planes.
- The other two projections of the line are parallel to their respective axes and are shortened in length.

	Horizontal line	Level line	Profile line
Solid and its three views			
Projection on Axonometric views			
Lines on the solid			

3-2 Projection of Lines



Is there a special position if a line is parallel to a plane?



3-2 Projection of Lines



(II) Perpendicular to Projection Plane

Projection characteristics:

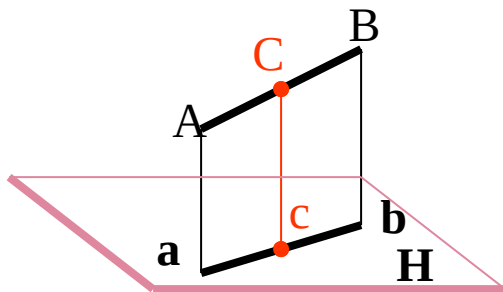
- The projection on the plane it is perpendicular to converges to a single point;
- The other two projections reflect the true length and are perpendicular to their respective axes.

	Horizontal line	Level line	Profile line
Solid and its three views			
Projection Axonometric views			
Lines on the solid			

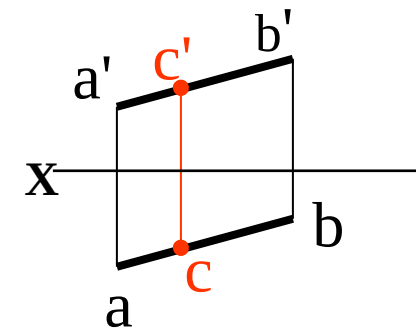
4. Relative Positional Relationship between Points and Lines

Relative position of point to line: **Point on line, Point off line**

Property: If a point is on a line, then each projection of the point must lie on the corresponding projection of the line.



Proportionality: The projections of the line are divided into the same spatial ratio.



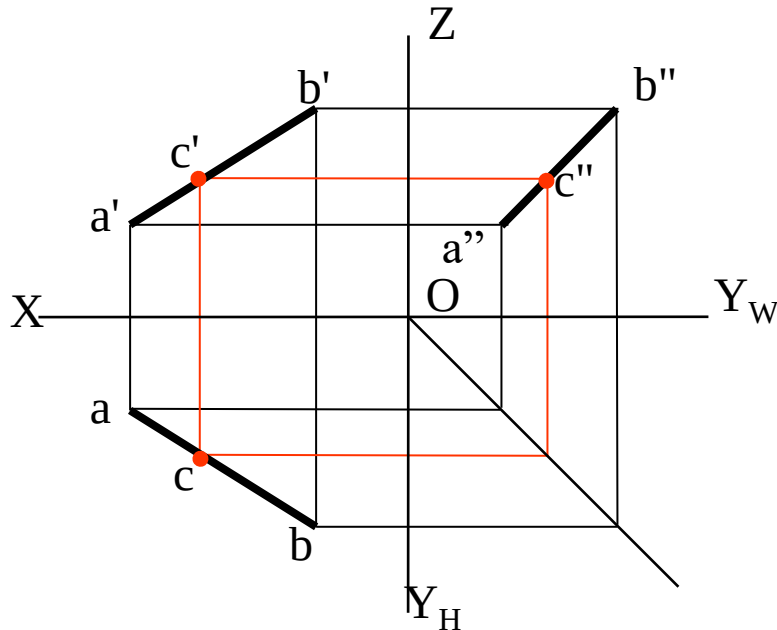
3-2 Projection of Lines



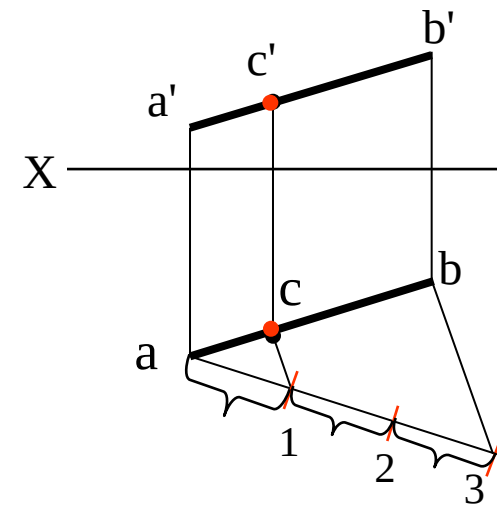
4. Relative Position of a Point to a Line

Point relative to line: **On the line, Off the line**

Example 4: Given that C is on AB, determine a and b based on c.

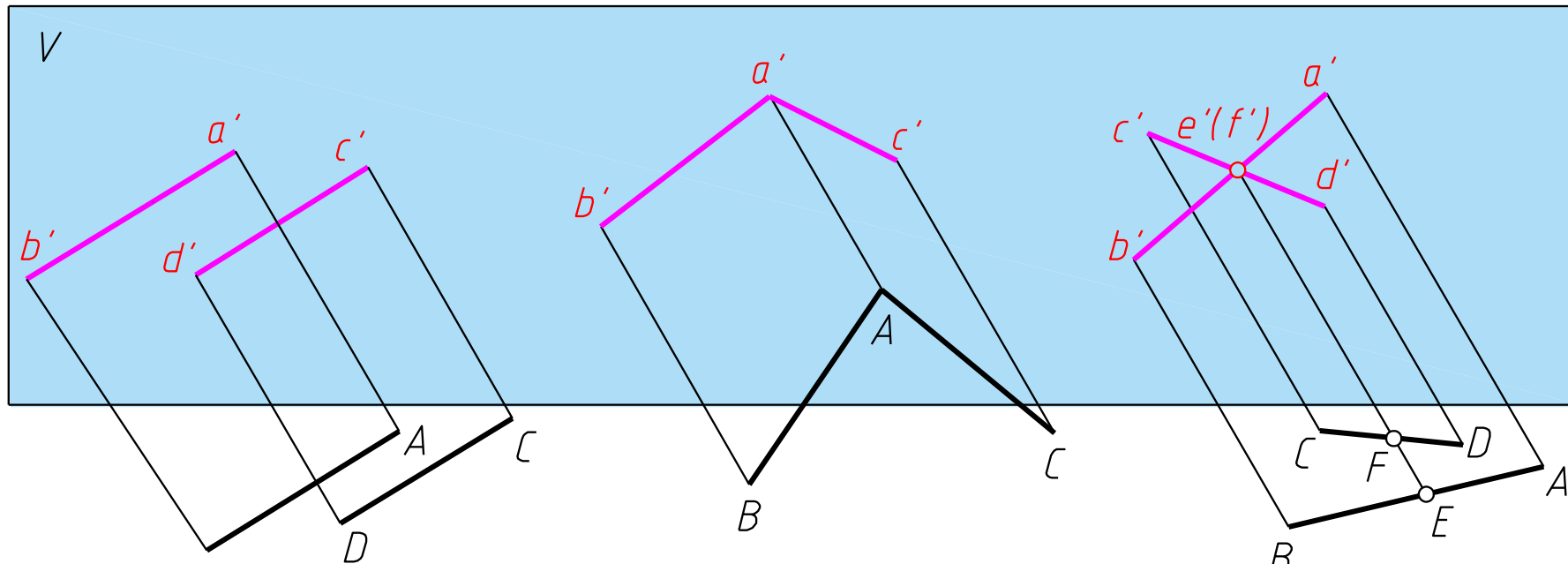


Example: Example 5: Given that C is on AB such that $AC:CB = 1:2$.



4. Relative Positional Relationship between Two Lines

Relative positions of two lines: **Parallel**, **Intersecting**, **Skew**



- **Parallel lines**

Intersecting lines

**Intersecting
lines**

3-2 Projection of Lines



4. Relative Positional Relationship between Two Lines

Two lines are parallel:

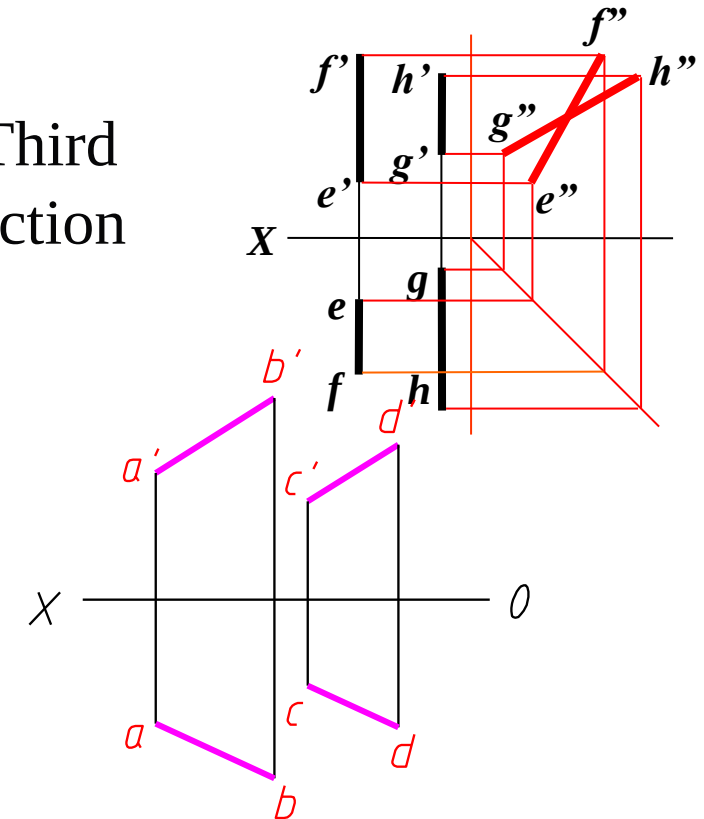
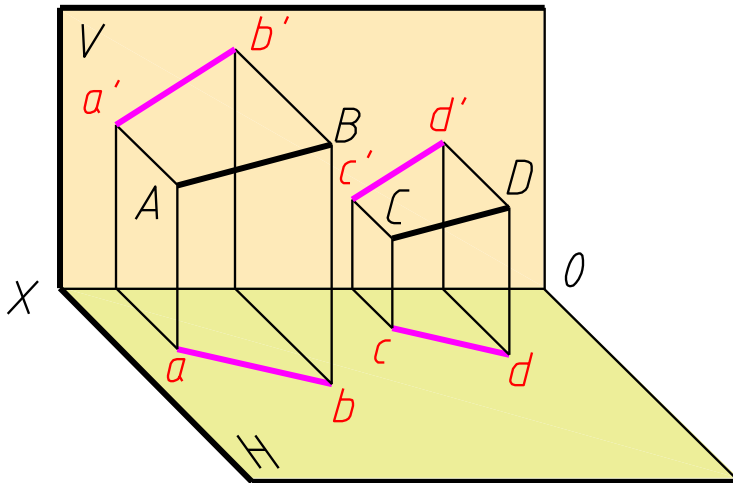
Spatial position: Coplanar, parallel

Projection pattern: Parallelism, proportionality

Judgment rule: Projections on the same plane are parallel

Example: Given that $ab//cd$ $a'b'//c'd'$ $ef//gh$ $e'f'//g'h'$, determine if lines AB and CD, EF and GH are parallel.

Method: Third plane projection



3-2 Projection of Lines



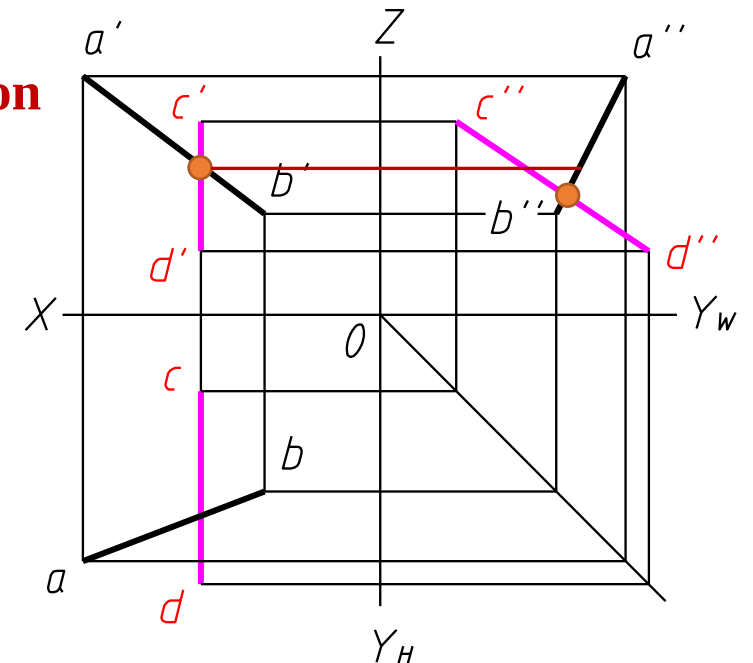
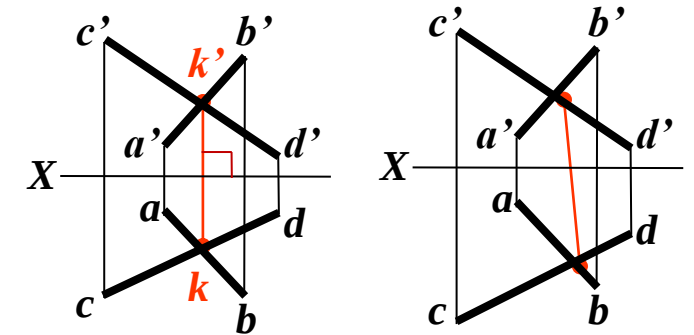
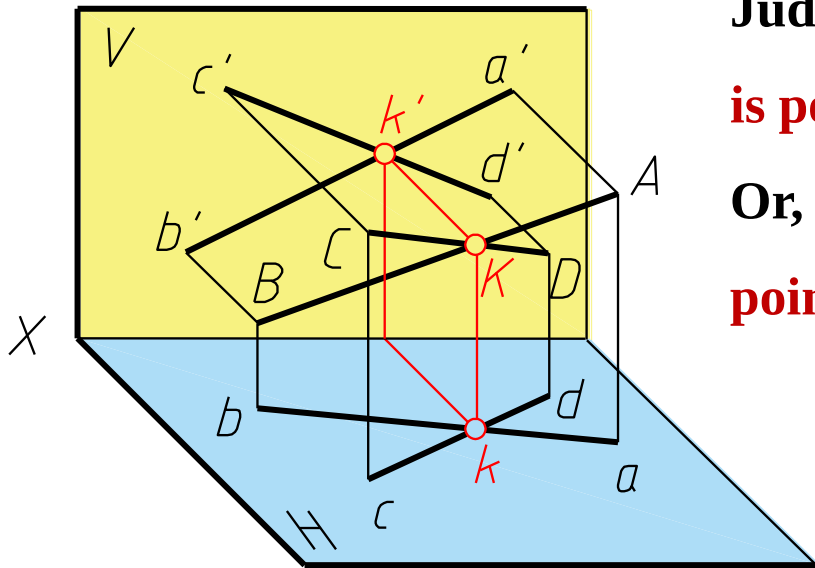
4. Relative Positional Relationship between Two Lines

(2) Two lines intersect:

Spatial position: There must be one intersection point in space.

Projection rule: The intersection point is a common point.

Judgment rule: The projection intersection is perpendicular to the projection axis.
Or, whether the three-plane projection points correspond.



3-2 Projection of Lines

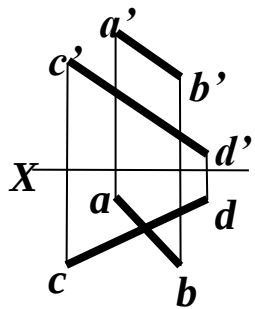
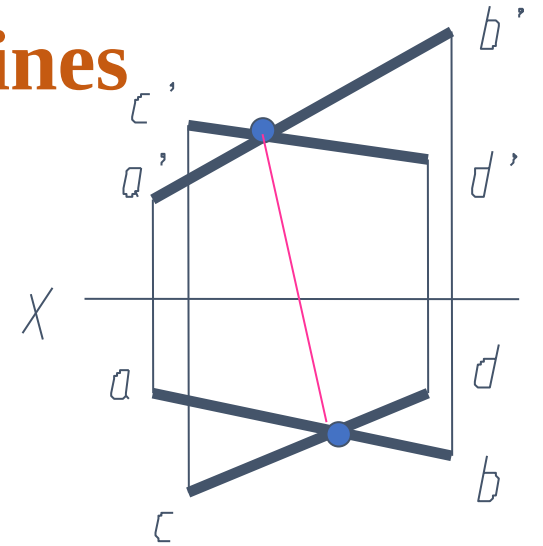
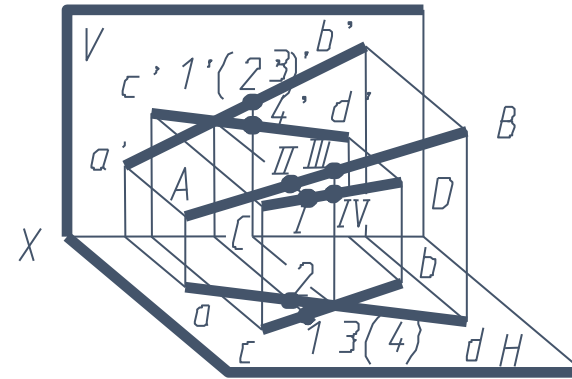


4. Relative Positional Relationship between Two Lines

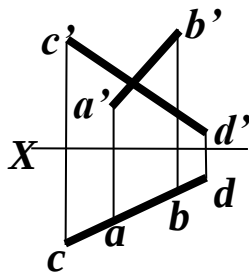
(3) Two lines are skew:

Spatial position: Not parallel, not intersecting.

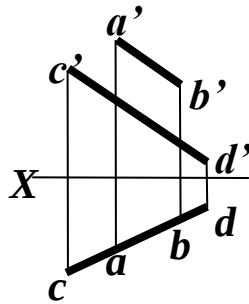
Judgment rule: The two lines are not coplanar.



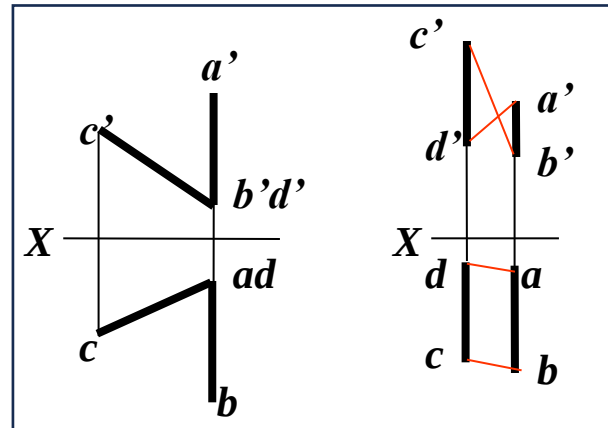
Skew



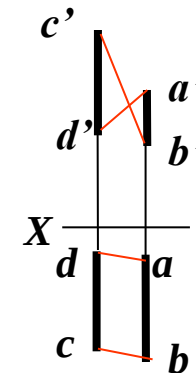
Intersecting



Parallel



Skew



Skew

Smoke
screen

Chapter 3 Projections of Points, Lines, and Planes

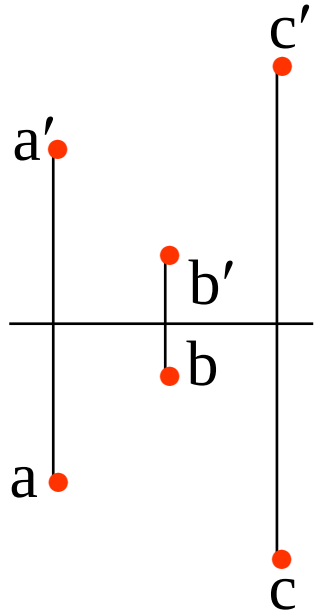
- **3-1 Projection of Points**
- **3-2 Projection of Lines**
- **3-3 Projection of Planes**

3-3 Projection of Planes

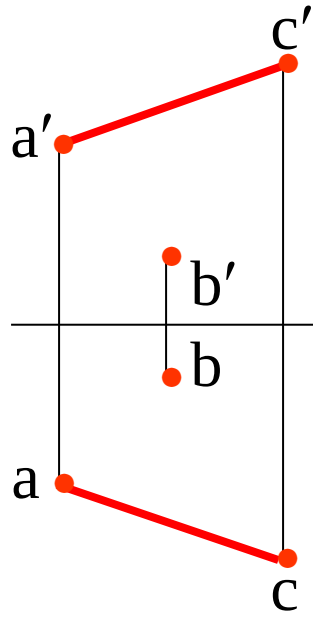


1. Methods of Representing a Plane

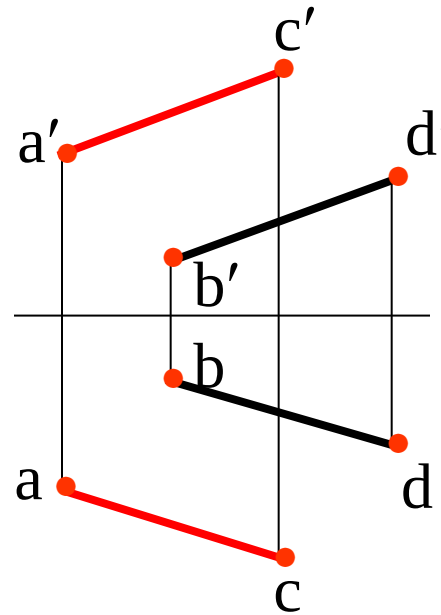
A plane is defined by a set of geometric elements, and there are several common representation methods:



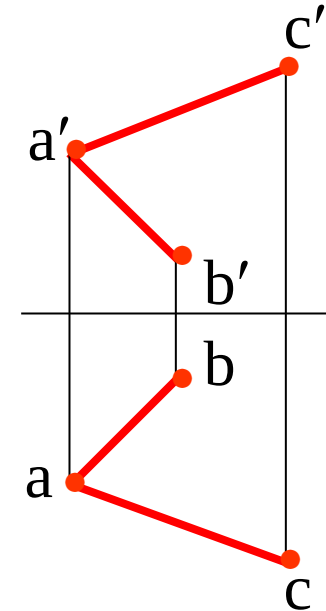
(1) Three non-collinear points



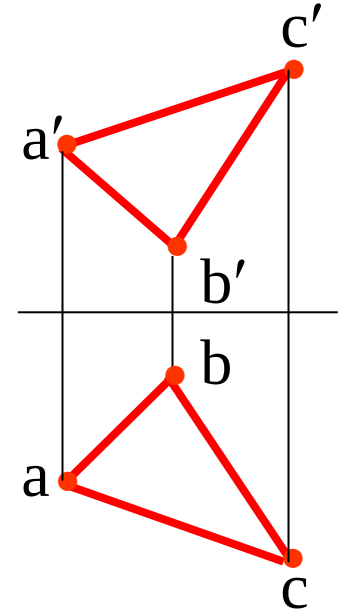
(2) A line and a point not on the line



(3) Two parallel lines



(4) Two intersecting lines



(5) Planar figure

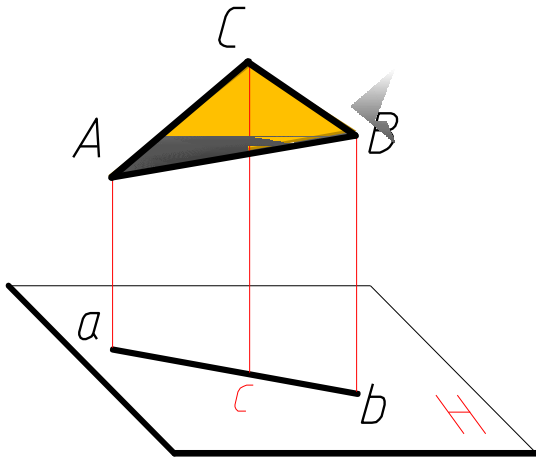
3-3 Projection of Planes



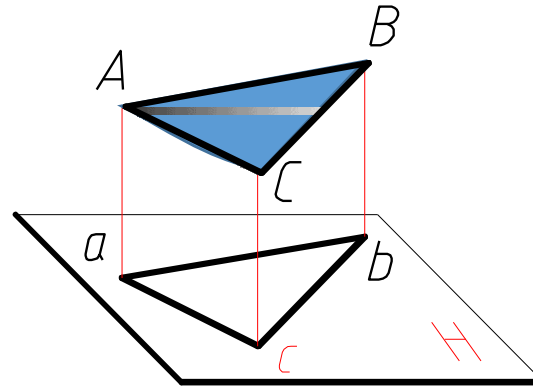
1. Projection Characteristics of a Plane

Relative position of a plane to the projection plane: Perpendicular, Parallel, Inclined

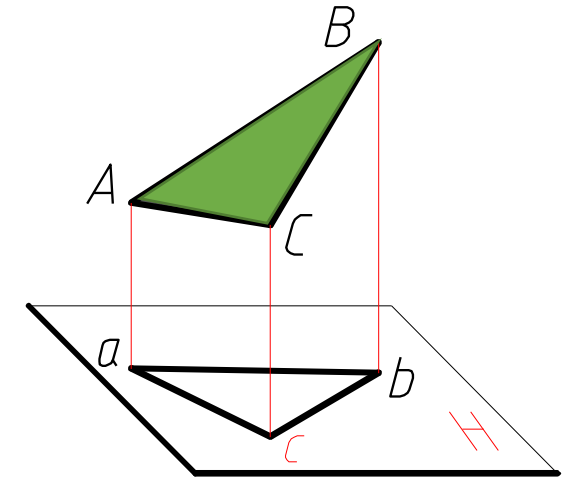
Perpendicular:
Gathering property



Parallel:
True shape property



Inclined:
Similarity property



Parallel: The projection reflects the true shape (**True shape property**) .

Vertical: The projection gathers into a single line (**Gathering property**) .

Inclined: The projection is similar to the original figure (**Similarity property**) .

3-3 Projection of Planes

1. Projection Characteristics of a Plane

The plane's position relative to the three projection planes is categorized into three types:

Perpendicular to one
projection plane, inclined
to the other two.

**Perpendicular to
Projection Plane**

{ Perpendicular to the Frontal Plane
Plumb vertical plane
Perpendicular to the Profile Plane

Special position plane

Parallel to one projection
plane, perpendicular to the
other two.

Parallel to Projection Plane

{ Frontal Plane
Horizontal Plane
Profile Plane

Inclined to all three
projection planes.

General Position Plane

3-3 Projection of Planes



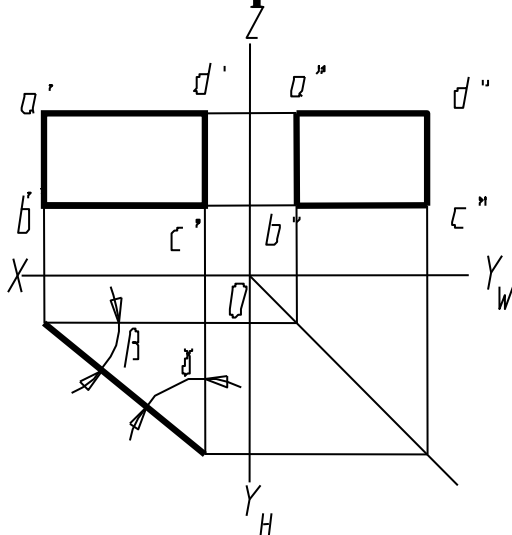
1. Projection Characteristics of a Plane

Position of a plane relative to a projection plane: **Perpendicular**

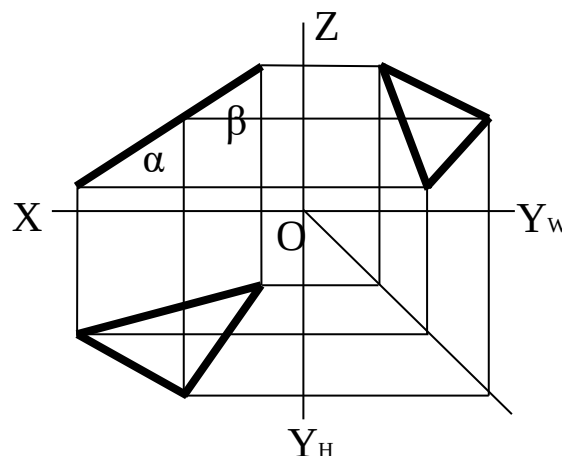
(1) Spatial position: **Perpendicular to a certain projection plane**

(2) Projection characteristics: $\left\{ \begin{array}{l} \text{Gathering property, Similarity} \\ \text{Reflects the angles with the other two principal projection planes} \end{array} \right.$

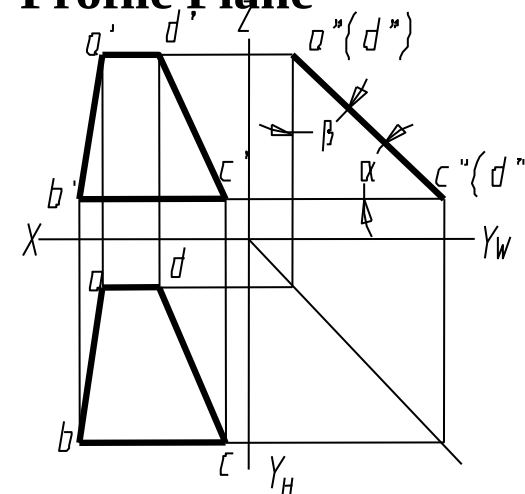
Plumb plane



Perpendicular to the Frontal Plane



Perpendicular to the Profile Plane



3-3 Projection of Planes

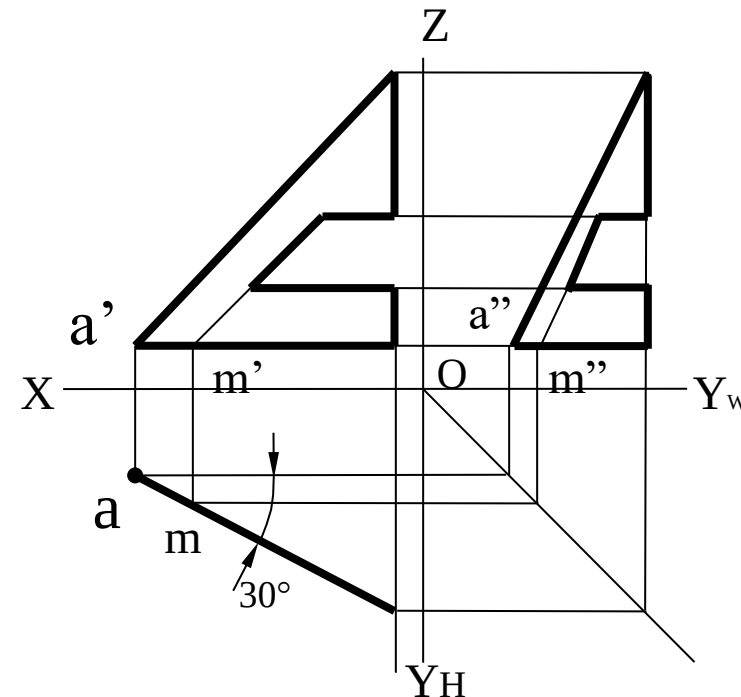


	Perpendicular to the Frontal Plane	Plumb plane	Perpendicular to the Profile Plane
Solid diagram			
Projection drawing			
Faces on solid			

3-3 Projection of Planes



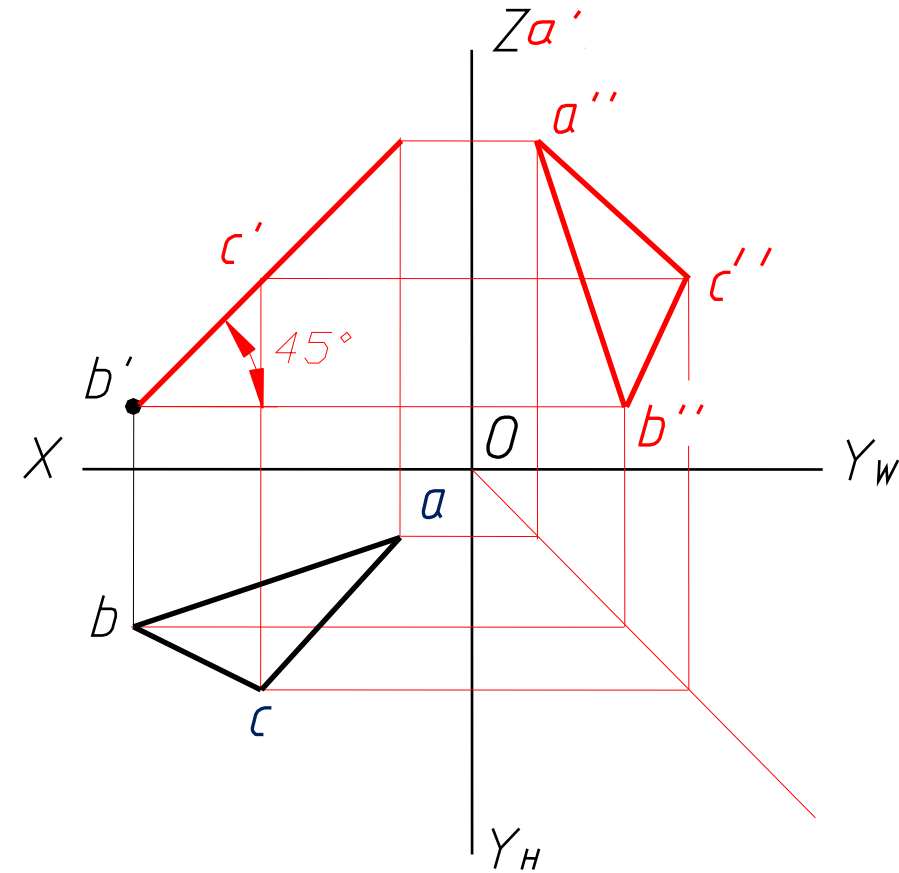
Example: Given a plane is a plumb vertical plane, with an angle of 30 degrees to the front vertical plane, and its front projection is known as 'a', find its horizontal projection and side projection.



3-3 Projection of Planes



Example: Plane $\triangle ABC$ is perpendicular to the V plane. The H plane projection is known as $\triangle abc$, and the V plane projection of point B is b' . The angle of inclination $\alpha=45^\circ$. Determine the V and W plane projections of the plane.



Analysis: Utilize the gathering property of the plane.

3-3 Projection of Planes

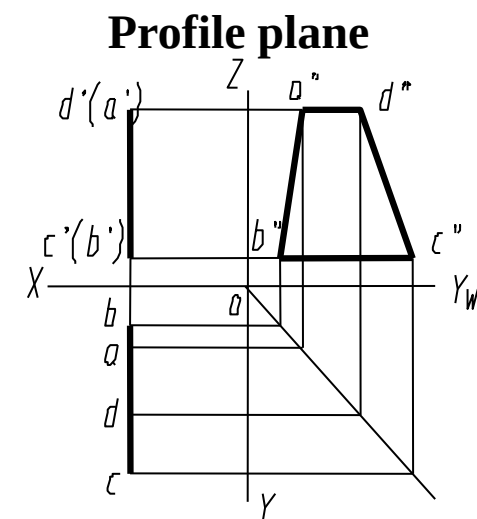
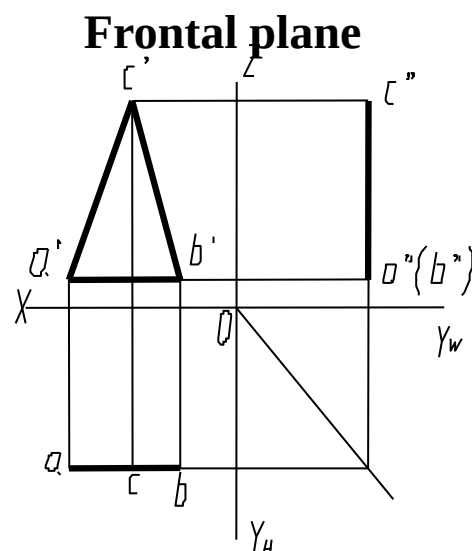
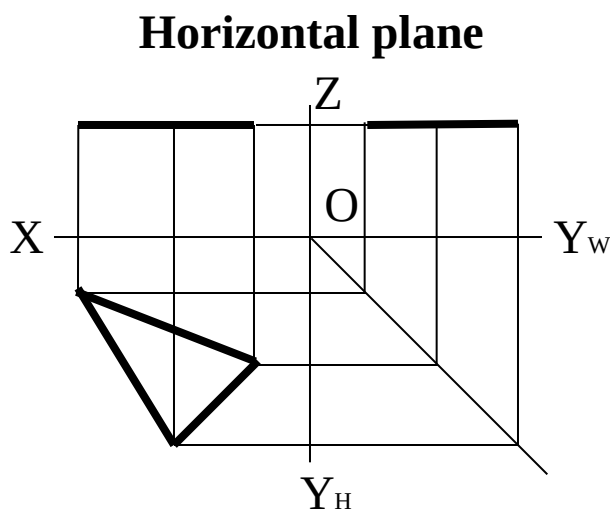


1. Projection Characteristics of a Plane

Position of a plane relative to the projection plane: **Parallel (a special case of perpendicular position)**

(1) Spatial position: **Parallel to a certain projection plane**

(2) Projection characteristics { **True shape, Gathering property**
 $\perp$ the other two projection planes, and the projection $\parallel$ the coordinate axis.



3-3 Projection of Planes



	Frontal plane	Horizontal plane	Profile plane
Solid diagram			
Projection drawing			
Faces on solid			

3-3 Projection of Planes



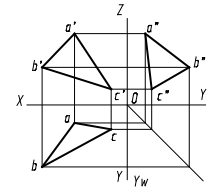
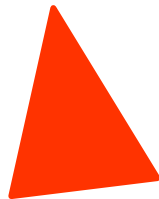
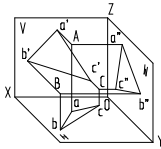
1. Projection Characteristics of a Plane

Position of a plane relative to the projection plane: **Inclined (general position)**

(1) Spatial position: **Inclined to all projection planes**

(2) Projection characteristics:

Similarity



3-4 Lines and Points within a Plane



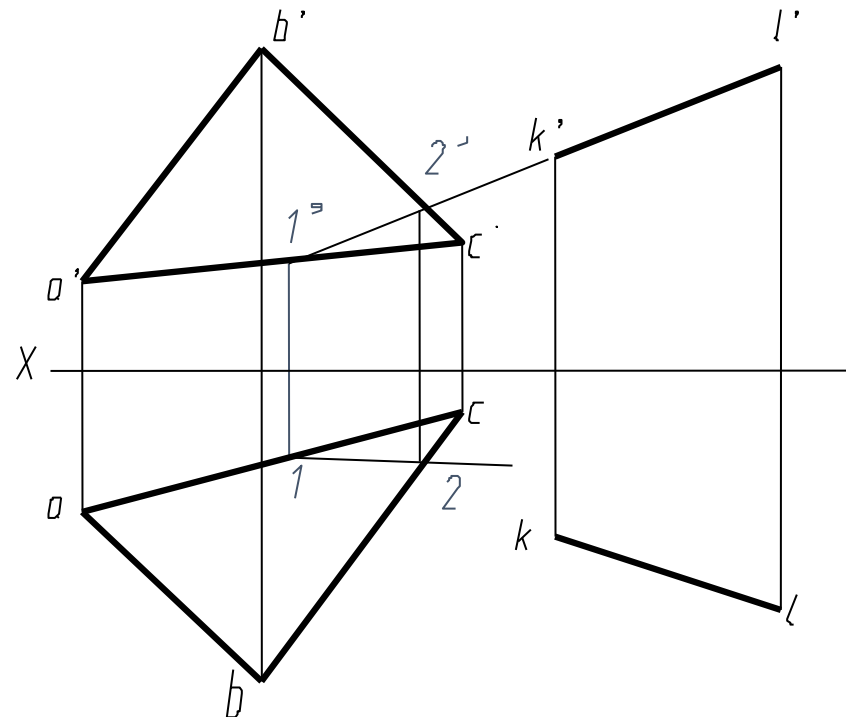
1. Geometric Conditions for a Line to be in a Plane

- (1) The **line passes through two points** within the plane.
- (2) It **passes through one point within the plane** and is **parallel to another line** within the plane.

➡ Basis for solving line drawing problems within a plane

Example:

Determine if line KL is within $\triangle ABC$.



2. Geometric Conditions for a Point to be in a Plane

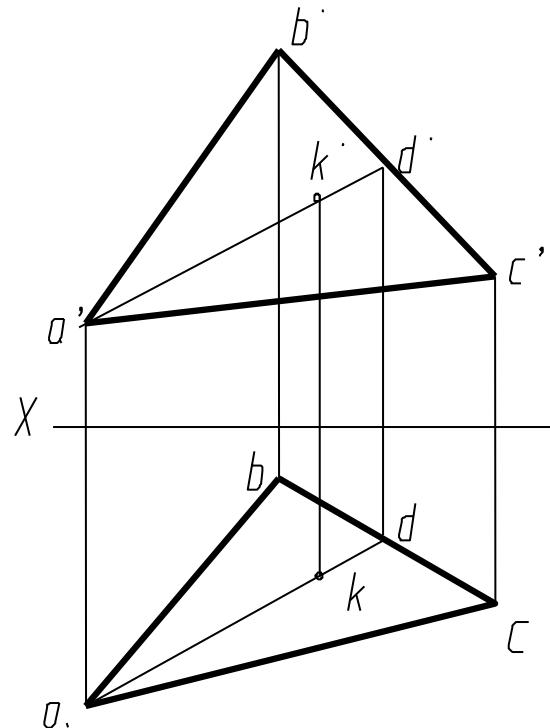
If a point is within a plane, then the point must **lie on a known line within that plane**.



Therefore, draw an **auxiliary line** within the plane and select a point on that line.

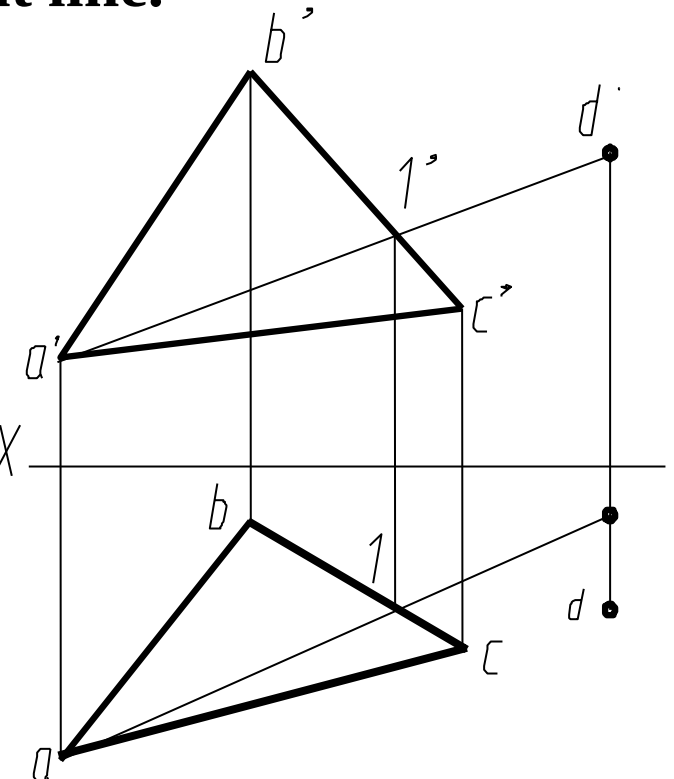
Example:

Find an arbitrary point K within ΔABC .



Example:

Determine if point D is within ΔABC .

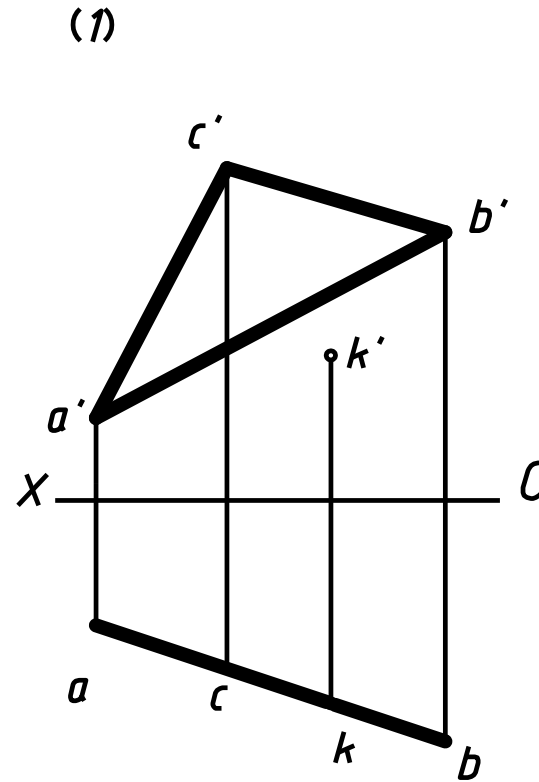


3-4 Lines and Points within a Plane

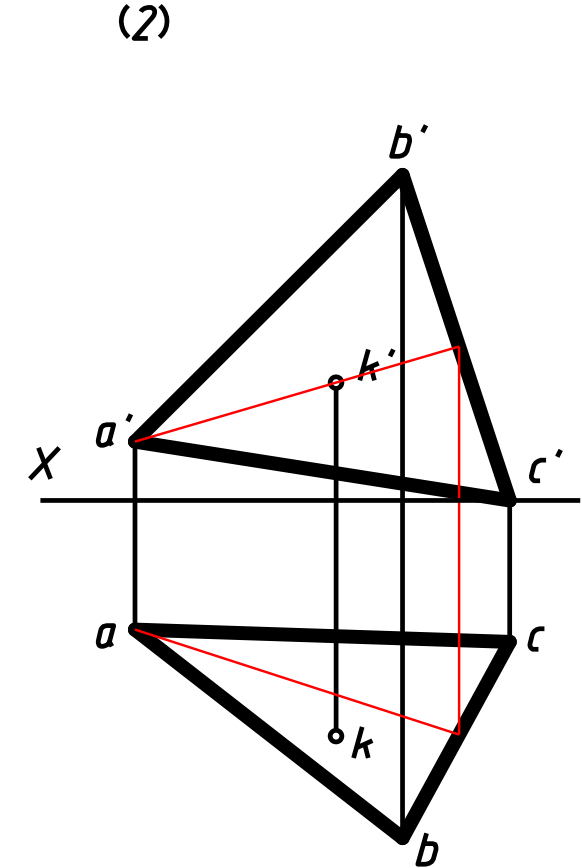


Example:

Given the two projections of a plane and point K, determine if point K is on the plane.



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3. Visibility of Line-Plane Intersection

Projection perpendicular line intersects with a general position plane

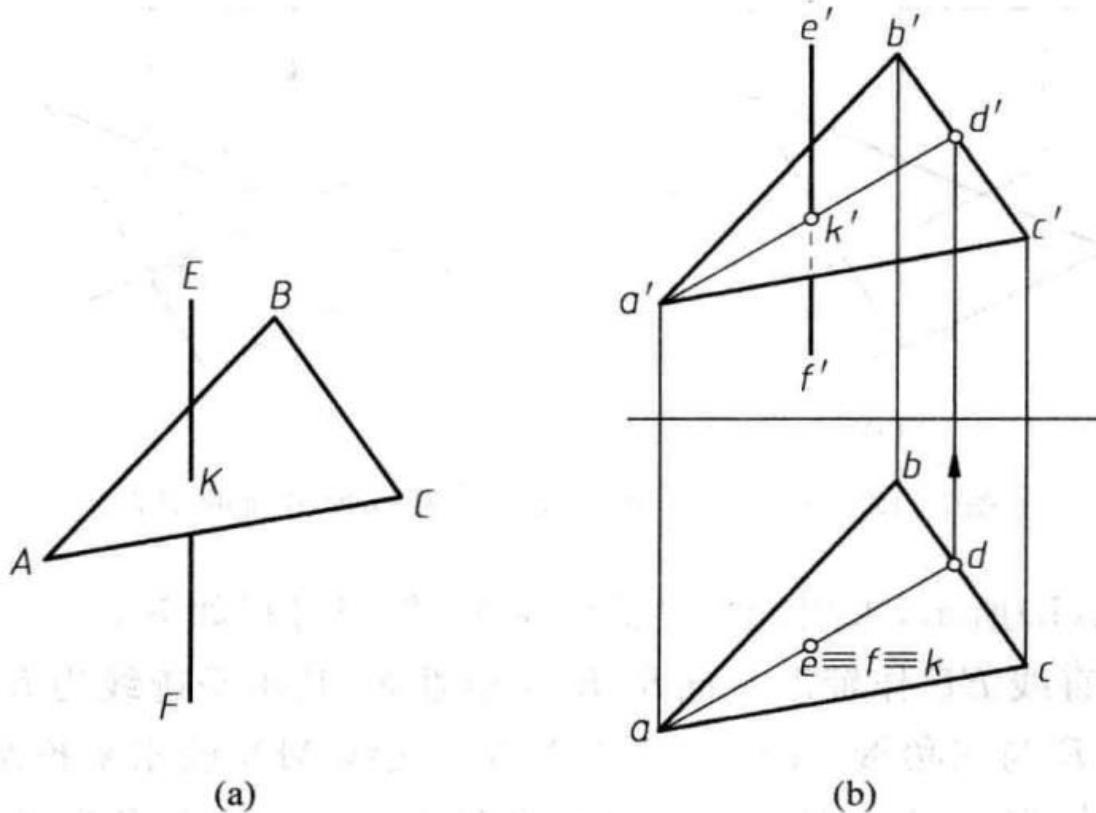


图 5-10 投影面垂直线与一般位置平面相交

Figure 5-10: Intersection of a Perpendicular Projection Line with a General Position Plane

- In the left diagram, line EF is perpendicular to the horizontal plane and intersects with $\triangle ABC$. How to determine the visibility (solid or dashed line) of the intersection between line EF and $\triangle ABC$ in the front plane?
- Determine the relative position of the line and plane in terms of up/down and front/back; **front and above are visible solid lines.**